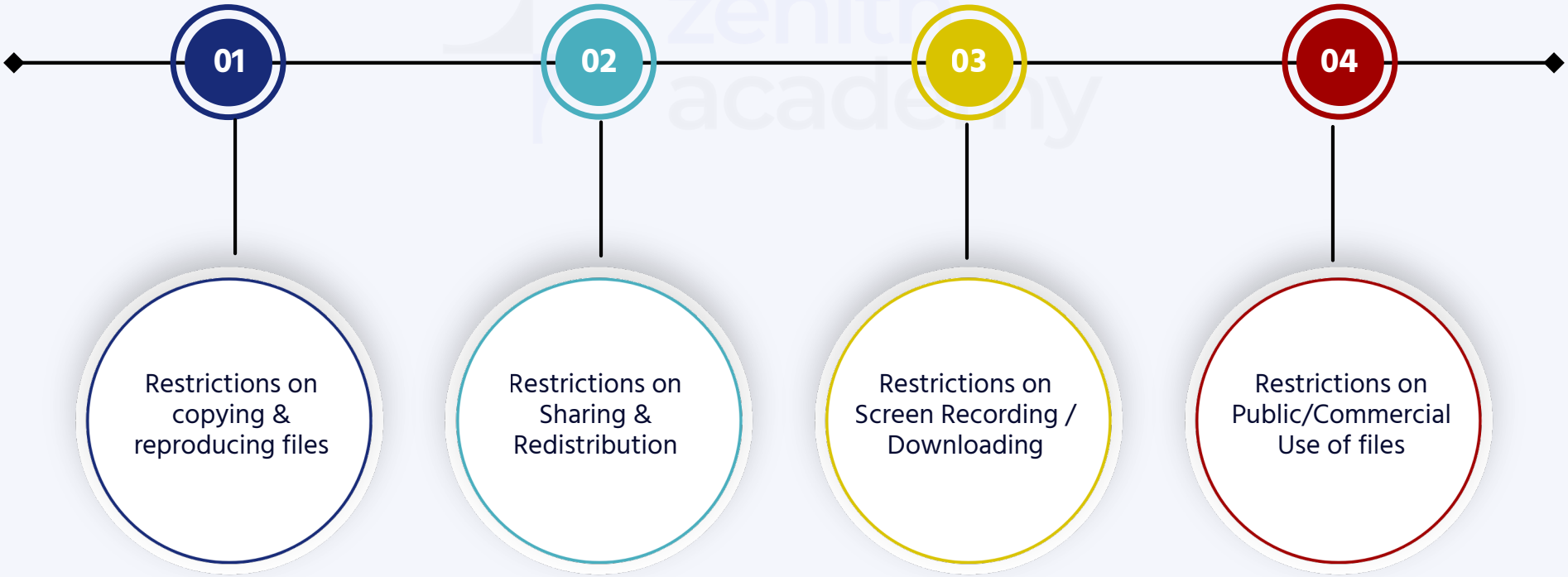
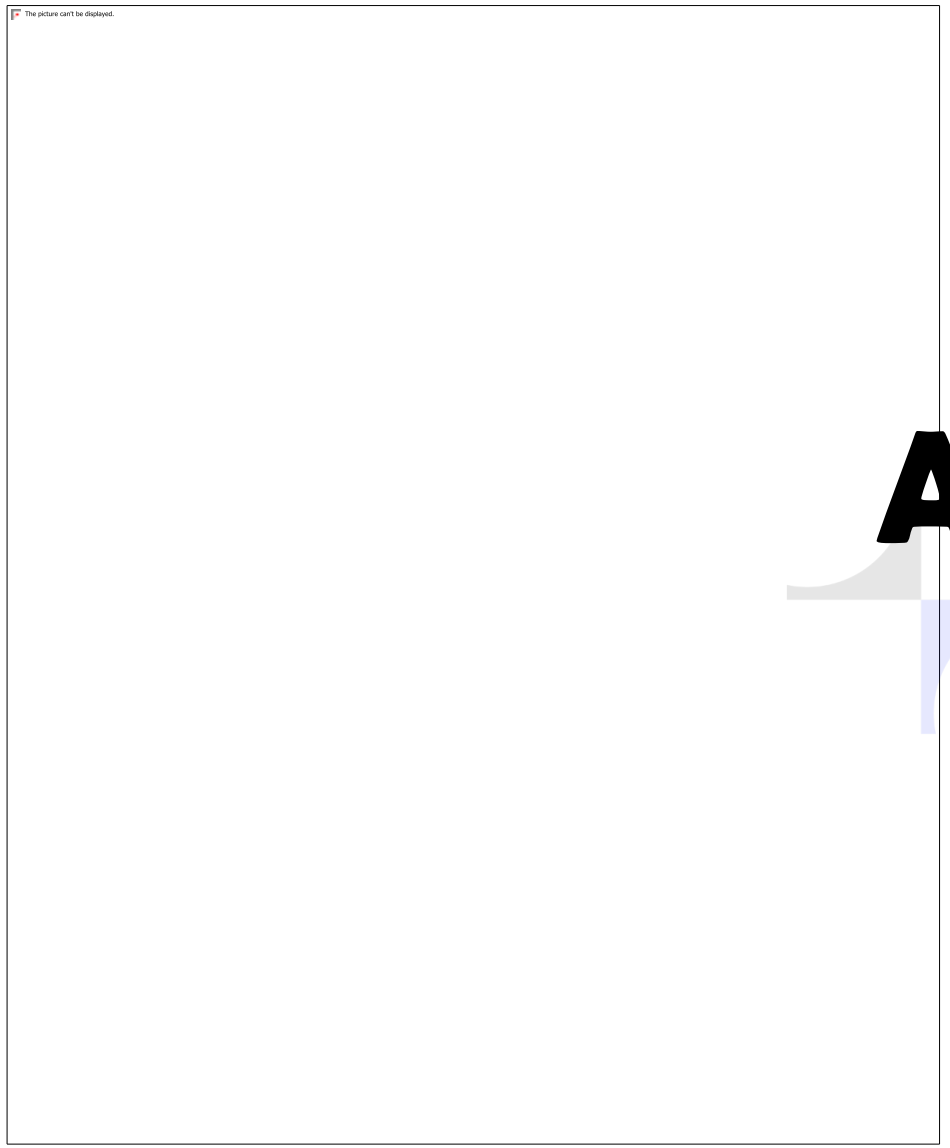
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Articular system

Dr. Kripa Krishnan

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Articulation : Union of any two or more bones of skeleton

Arthrology : Study of joint

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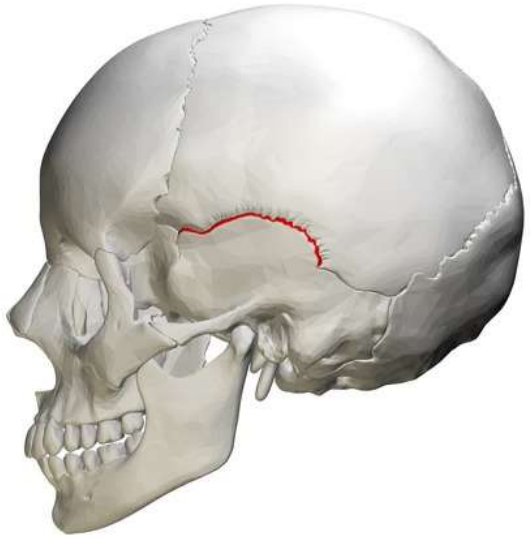
Classification of joints- Based on degree of motion

1. Synarthrosis / Fibrous / fixed joints / immovable joint
2. Amphiarthrosis / Cartilaginous /slightly movable joints
3. Diarthrosis / synovial joint / freely movable joints

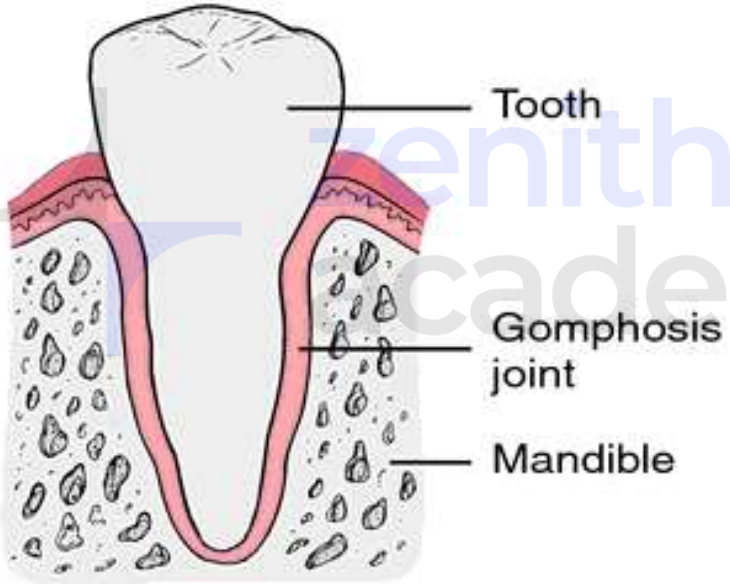
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1. IMMOVABLE JOINT

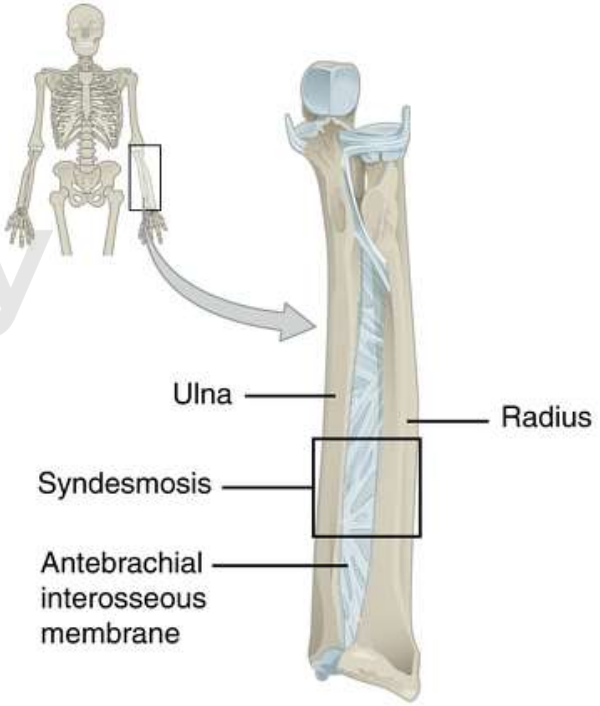
1. Suture



2. Peg / Socket joint/ Gomphosis :
Teeth in their sockets



3. Syndesmosis:
Articulating surfaces connected by membrane
Eg:-Radio-ulnar joint, Tibio-fibular joint

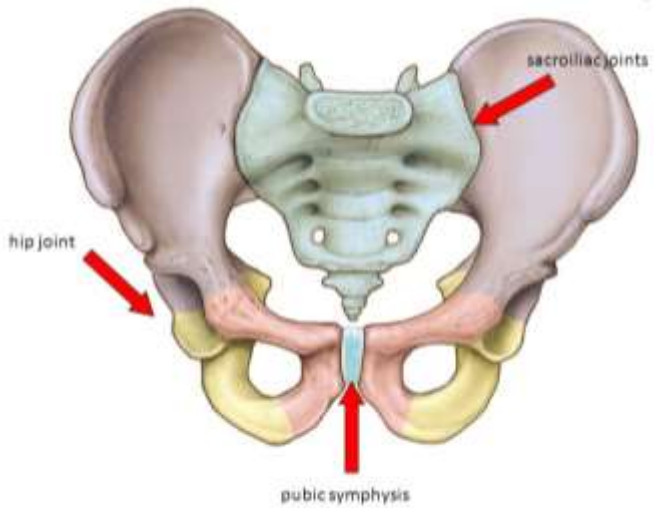


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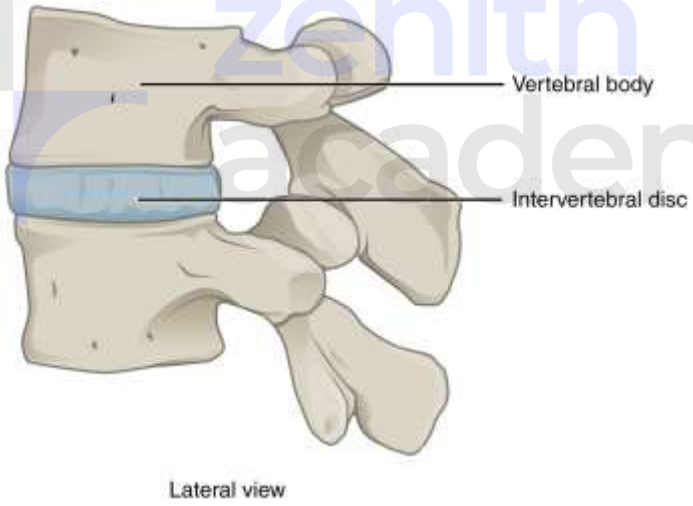
2. SLIGHTLY MOVABLE JOINTS

Joint surfaces are separated by some intervening substance/ cartilage.

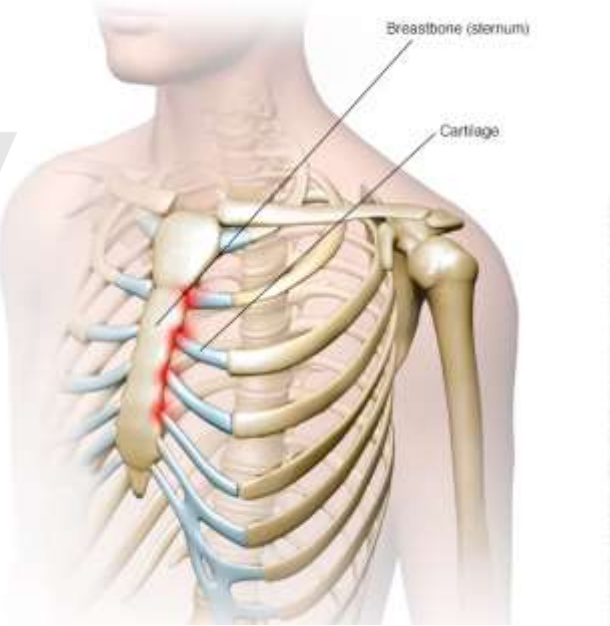
1. Pubic Symphysis



2. Inter vertebral joint



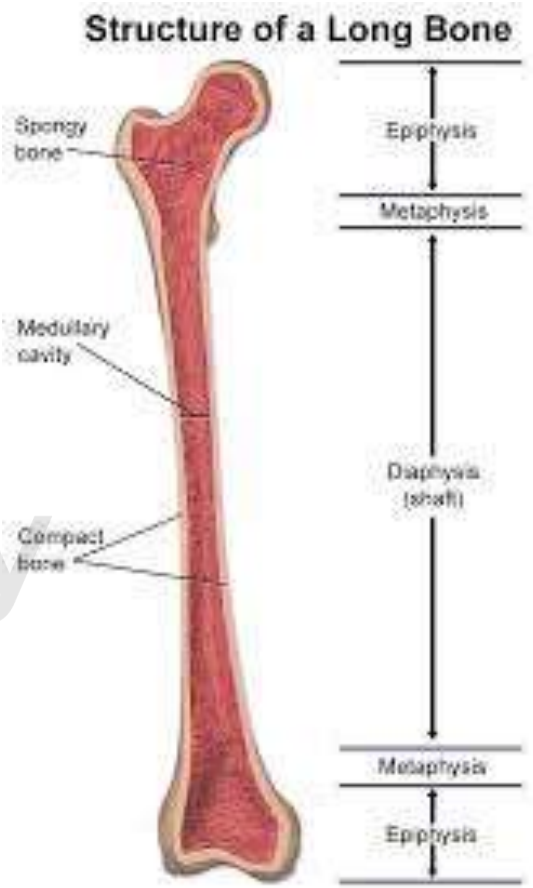
3. Joint between manubrium and body of sternum



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Temporary or primary cartilaginous joints are found between the diaphysis and epiphyses of the long bone.

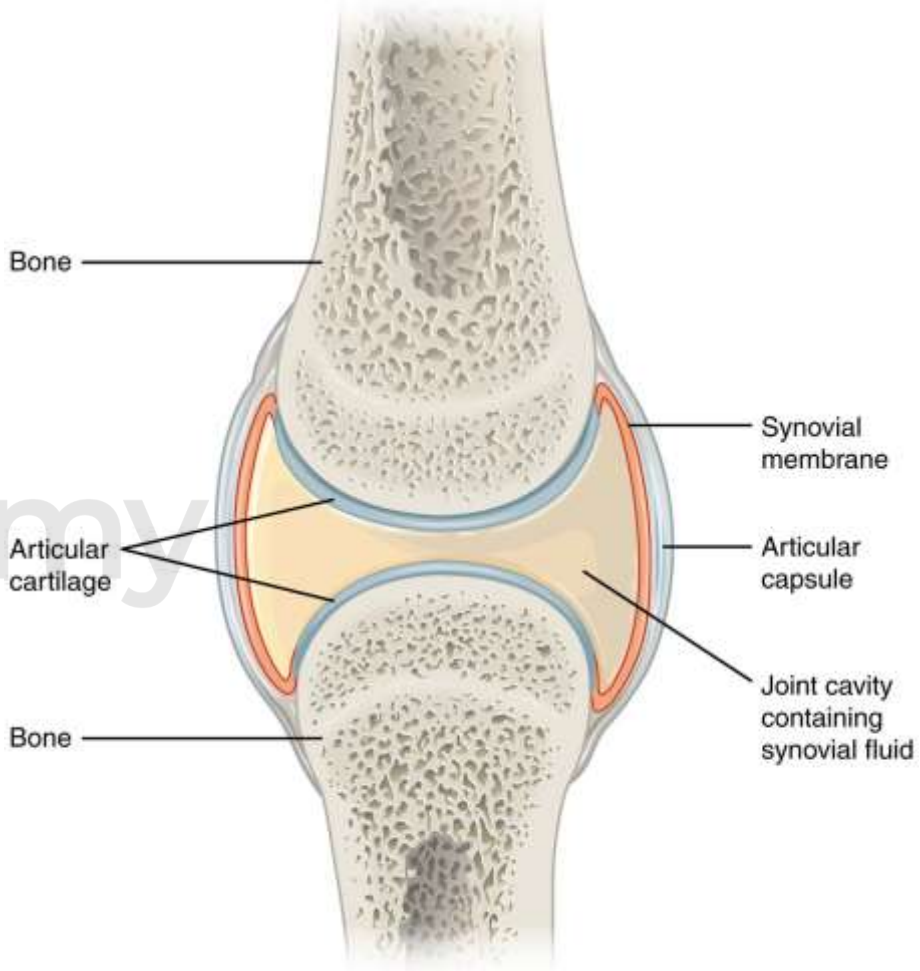
(helps in bone growth)



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3. SYNOVIAL JOINT/ FREELY MOVABLE JOINT

- Covered by hyaline cartilage
- Ligaments are required
- A joint cavity present: a cavity enclosed by capsule of fibrous tissue usually strengthened by ligament.
- There are 6 varieties.

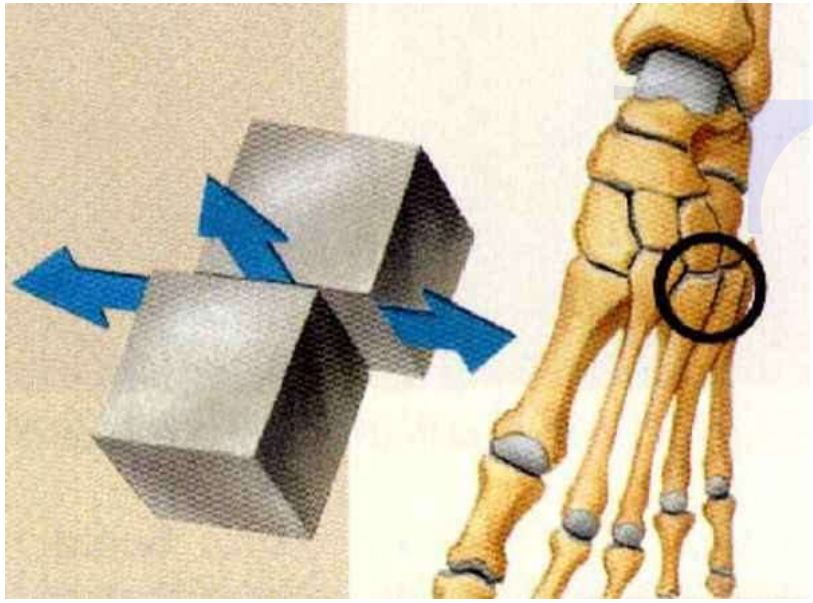


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1. Gliding joint / Plane joint

2 flat surface of bone glide on each other

Eg: Joints of carpal and tarsal

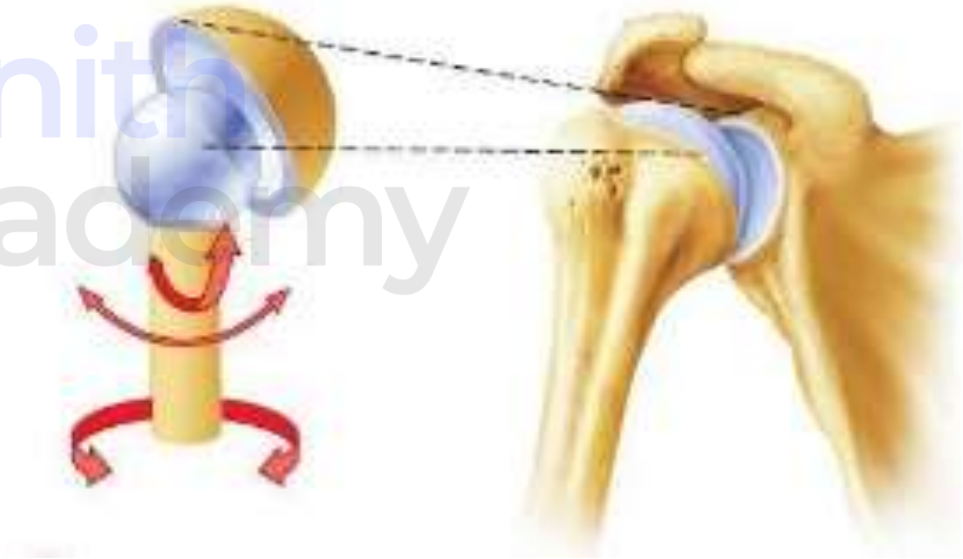


2. Ball and socket joint

One rounded extremity fits into a cavity of another bone.

Permitting movement in all directions.

Eg: Hip and Shoulder



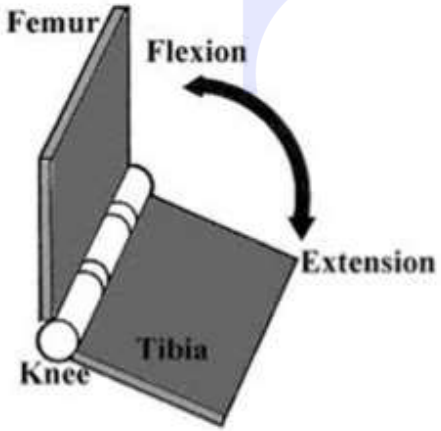
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3. Hinge joint

One rounded surface is received into another. Movement is only possible in one plane, as in hinge.

Eg: Elbow , Knee



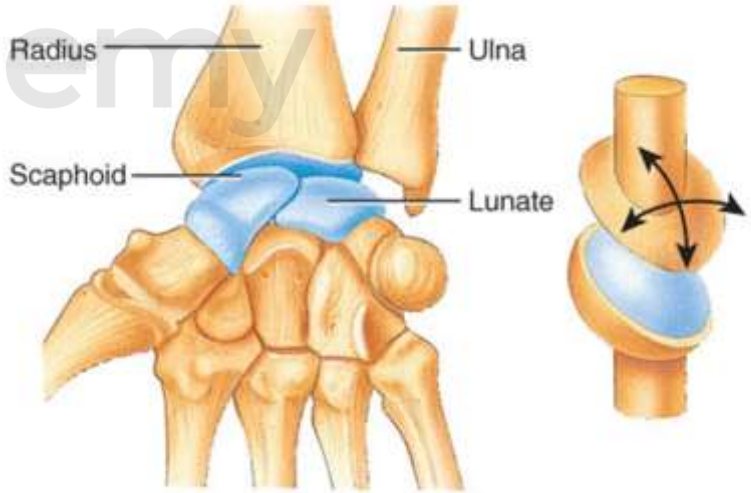


4. Condyloid joint

Similar to hinge joint, but permit movement in two planes, lateral as well as backward & forward.

Flexion & extension , Abduction &Adduction , sightly circumduction are possible, but not rotation.

Eg: Wrist



(d) Condyloid joint between radius and scaphoid and lunate bones of the carpus (wrist)

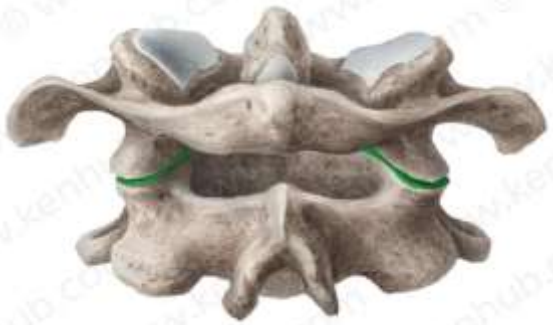
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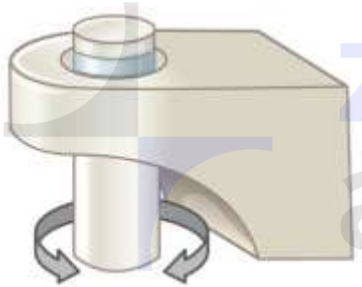
5. Pivot joint

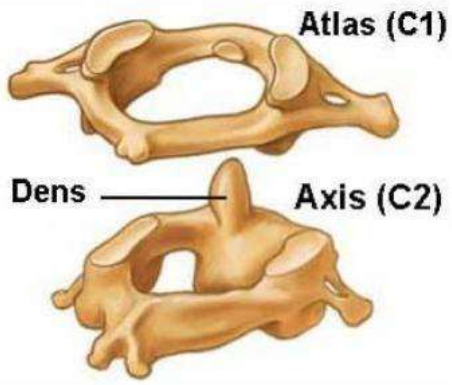
Only rotation is possible, as movement of head.

Eg : C1 rotates around C2

Movement of radius on ulna in pronation & supination





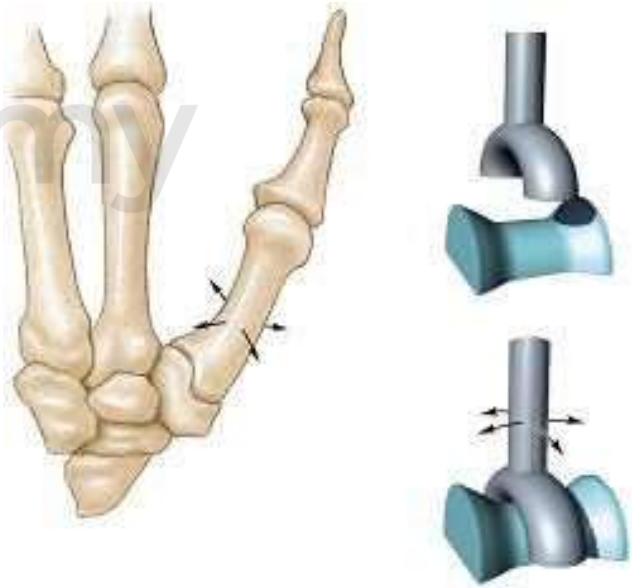




6. Saddle joint /Joint of reciprocal reception

Joint that allows for a wide range of movement.

Eg : between trapezium and first metacarpal bone of thumb. Enabling thumb to be opposed to the fingers.



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MOVEMENTS IN SKELETAL JOINTS
3 types

Gliding movement

Two flat surface move on each other

Angular movement

Described according to direction in which movement take place.

Flexion- extension

Abduction- adduction

Rotation movement

One bone moves around or within another bone.

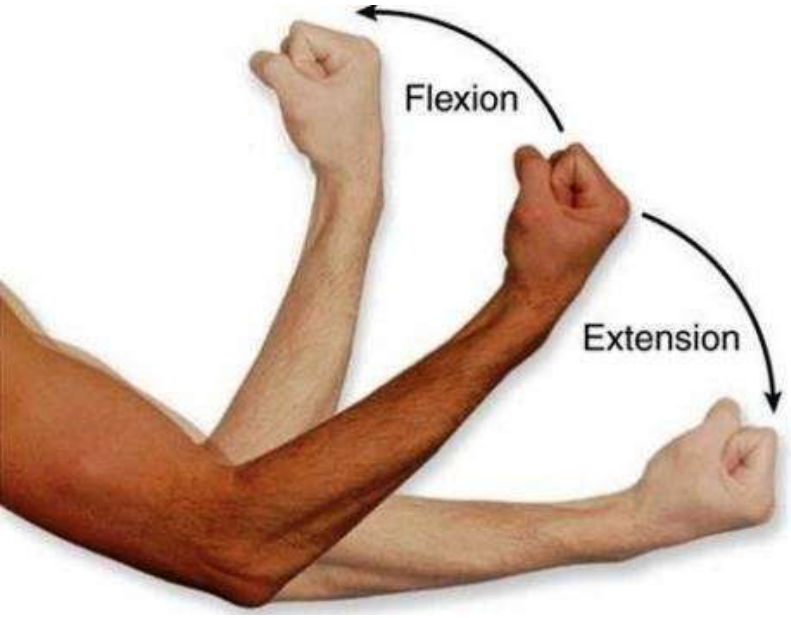
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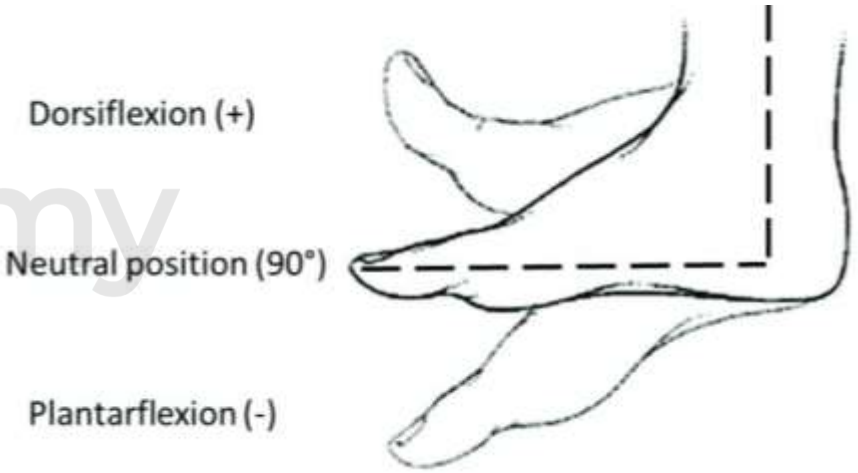
FLEXION – EXTENSION

Flexion : bending or doubling up

Extension : stretching or straitening out

Take place around an axis transversely placed.



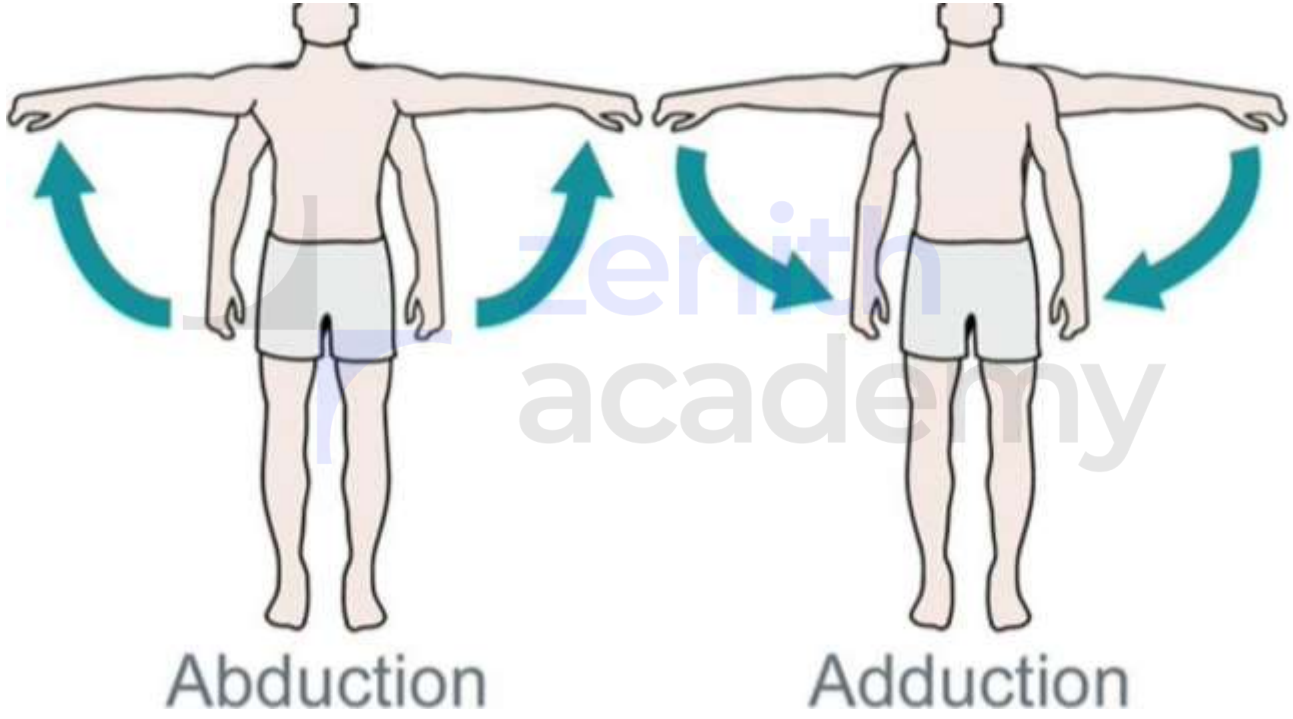


DORSI-FLEXION & PLANTAR FLEXION

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ABDUCTION & ADDUCTION

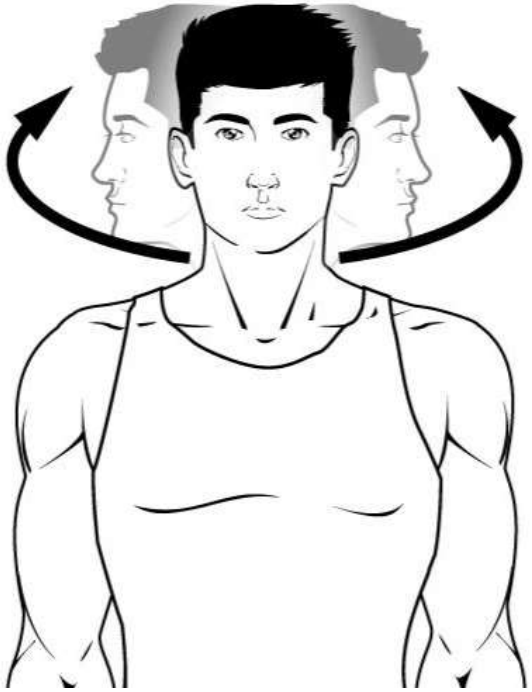
In a direction away from medial aspect.



Towards the medial aspect.

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ROTATION

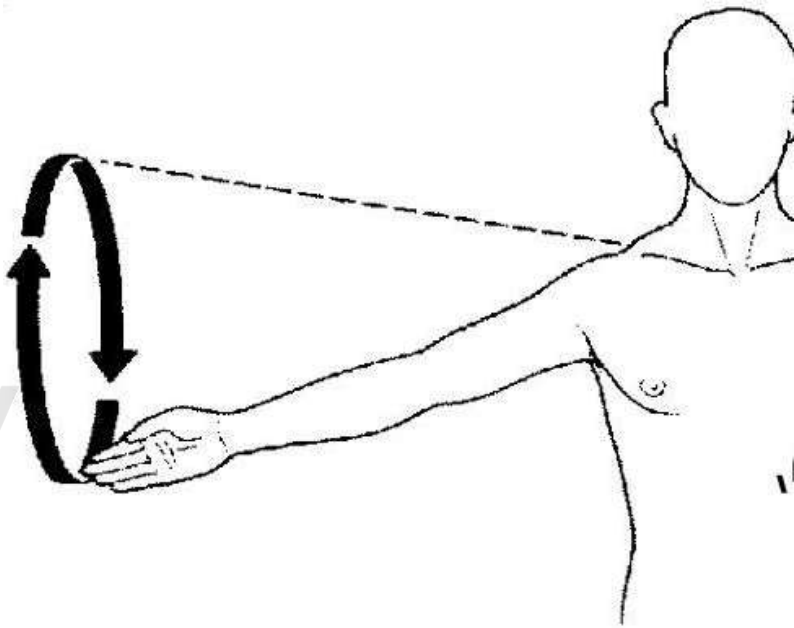


CIRCUMDUCTION

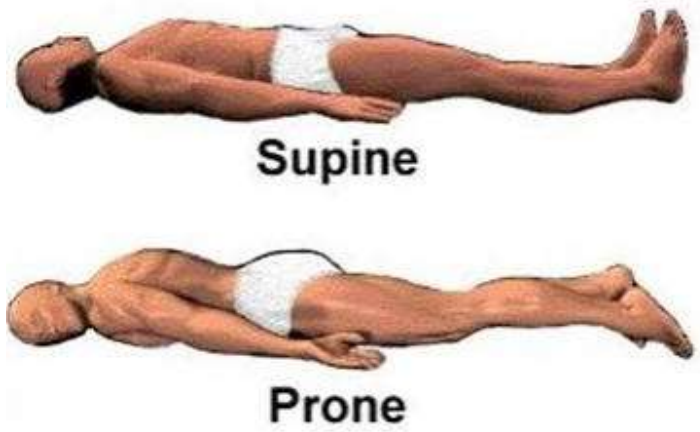
CIRCUMDUCTION = ROTATION + ANGULAR MOVEMENTS

This action includes

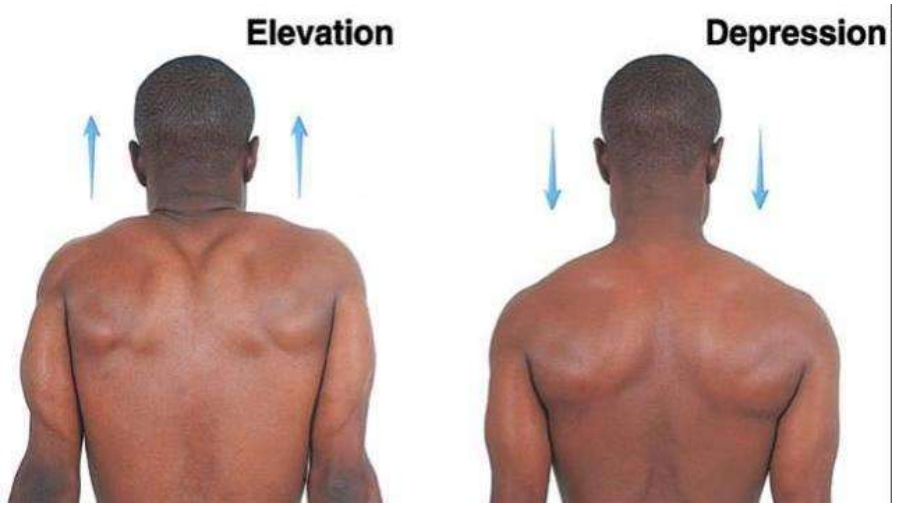
- Flexion
- Abduction
- Extension
- Adduction
- Some rotation



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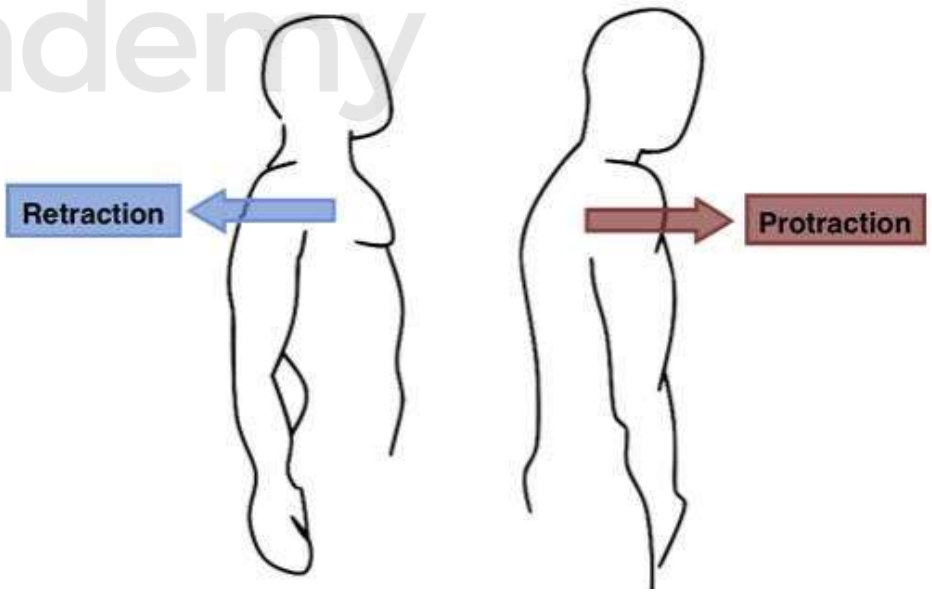




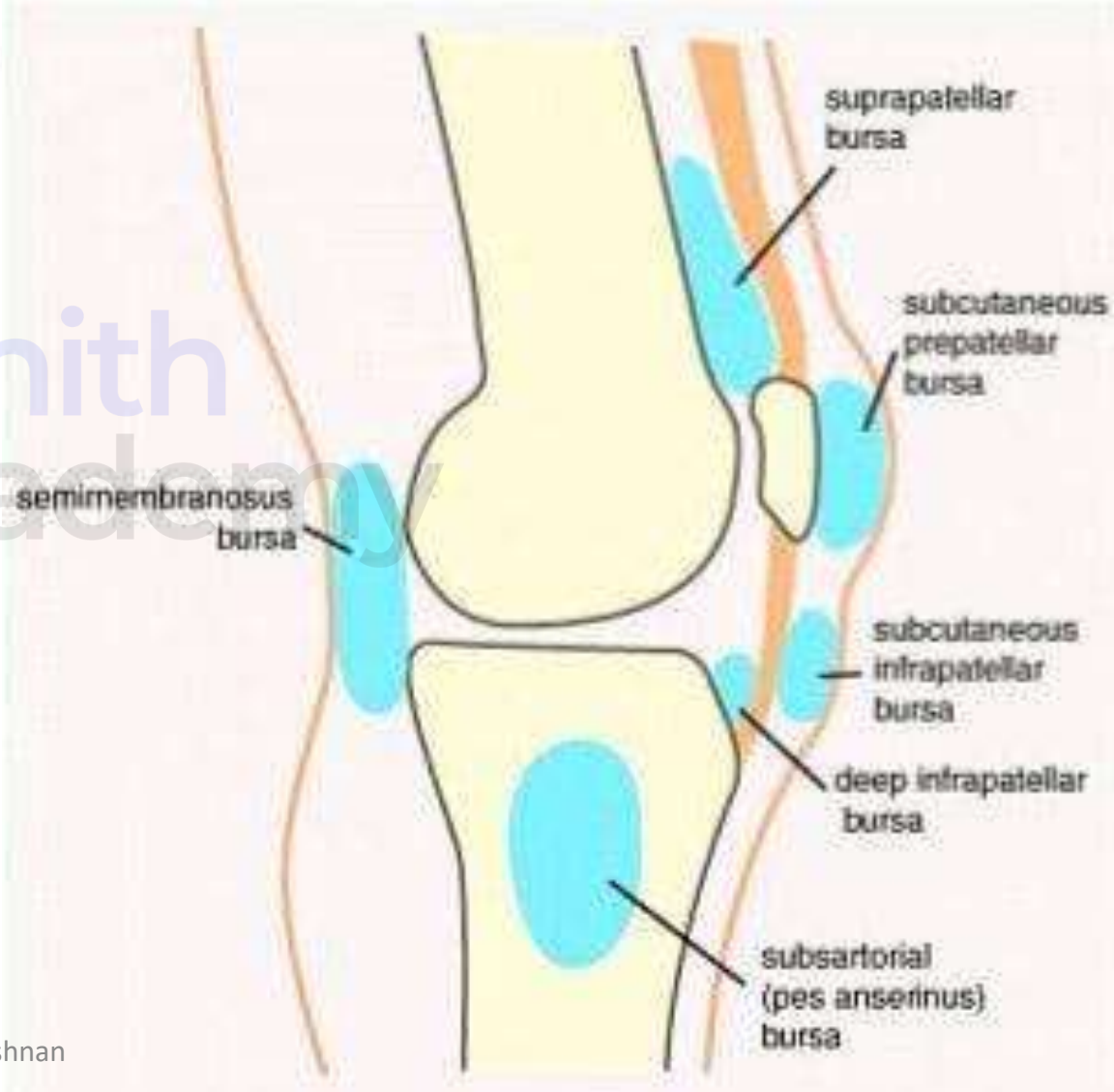
Opposition & Reposition


(b) Inversion


(c) Eversion

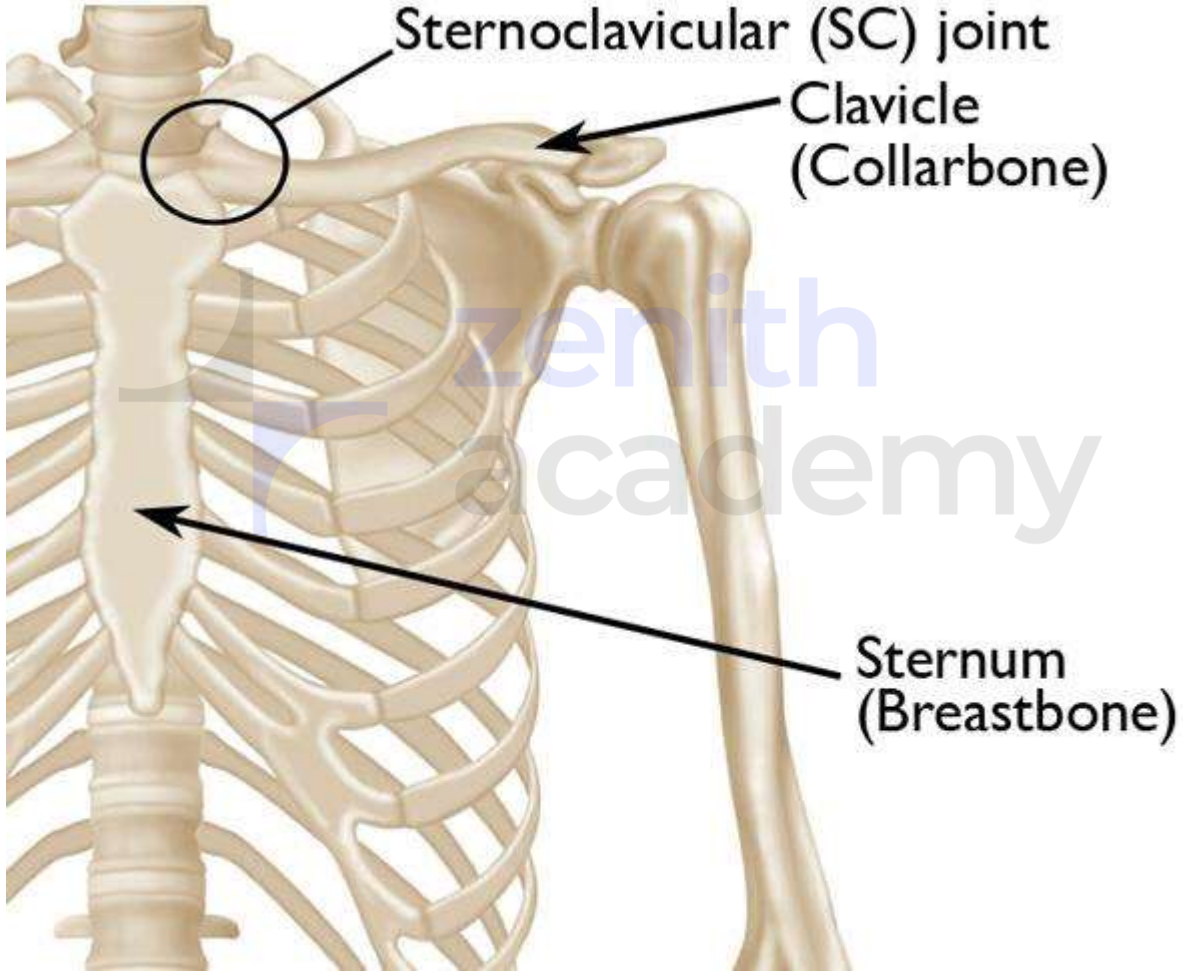


- To avoid friction during movement, the spaces between tendons, ligaments and bones have special structures with synovial membrane lining called **Bursae** (singular Bursa).
- Bursae are mainly of three types :
  1. Subfascial bursa
  2. Subcutaneous bursa
  3. Subtendinous bursa



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STERNO- CLAVICULAR JOINT

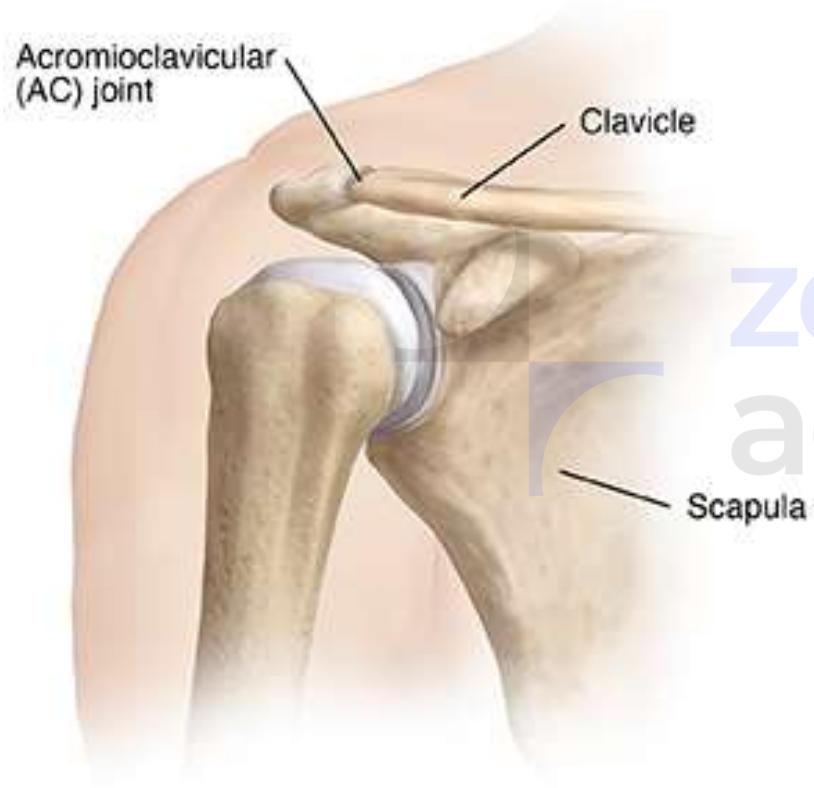


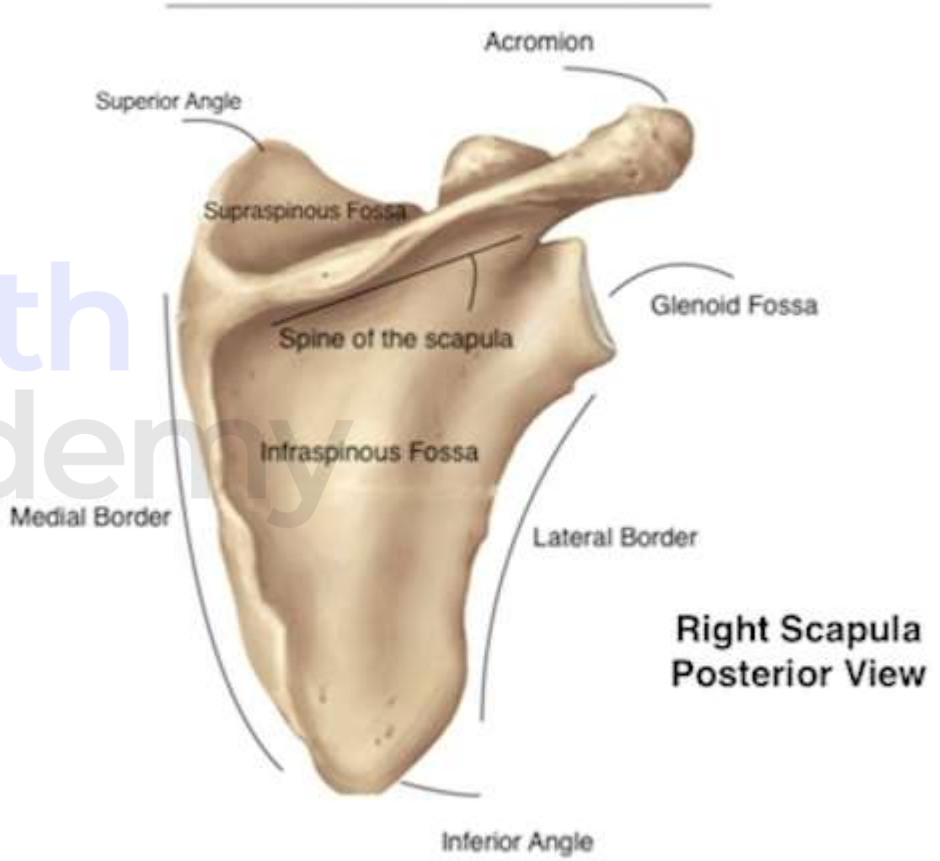
Gliding joint

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ACROMIO-CLAVICULAR JOINT

Gliding joint

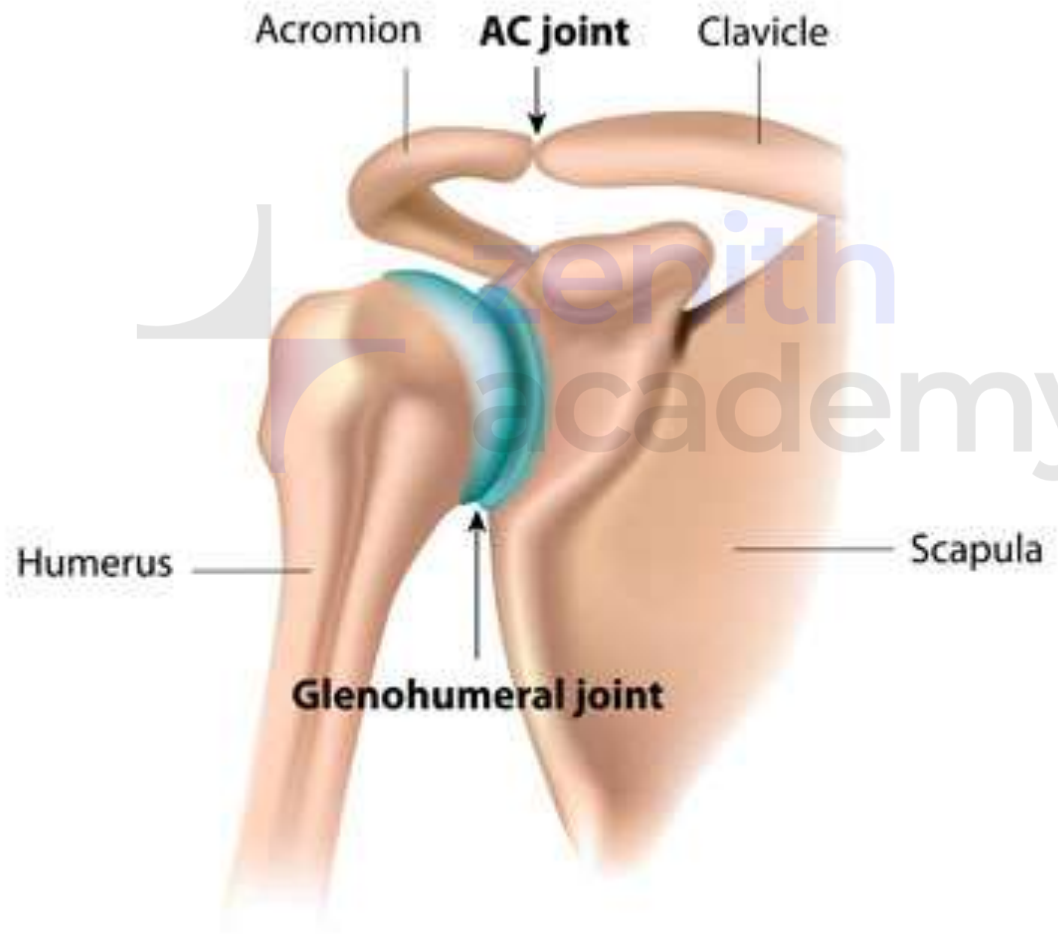




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SHOULDER JOINT

HUMERO-SCAPULAR JOINT



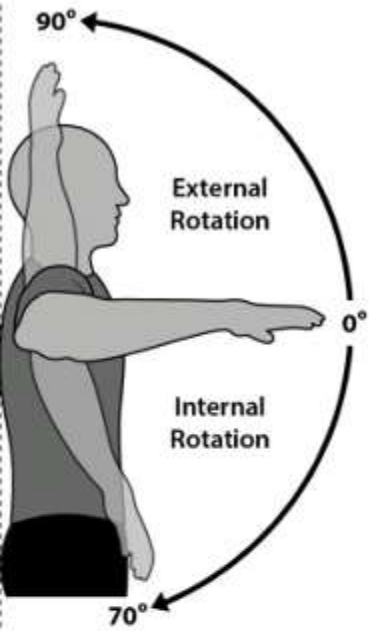
Synovial Joint

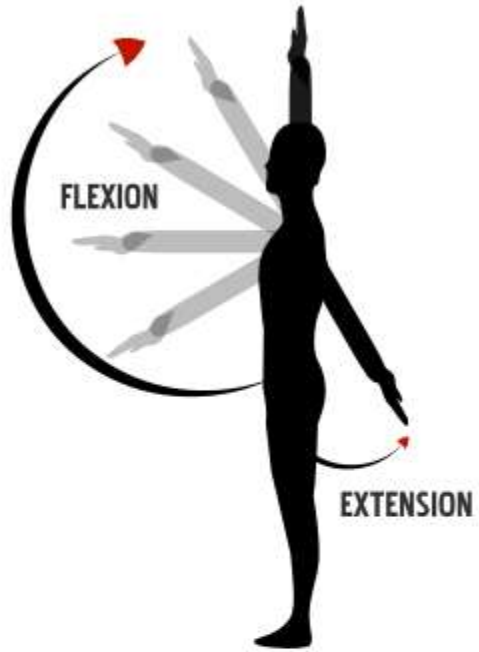
Movements

- In all direction
- Abduction
- Adduction
- Flexion
- Extension
- Medial &lateral rotation
- circumduction

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SHOULDER MOVEMENTS





Fitness Programer.Com

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HORIZONTAL FLEXION



HORIZONTAL EXTENSION



ABDUCTION



CIRCUMDUCTION



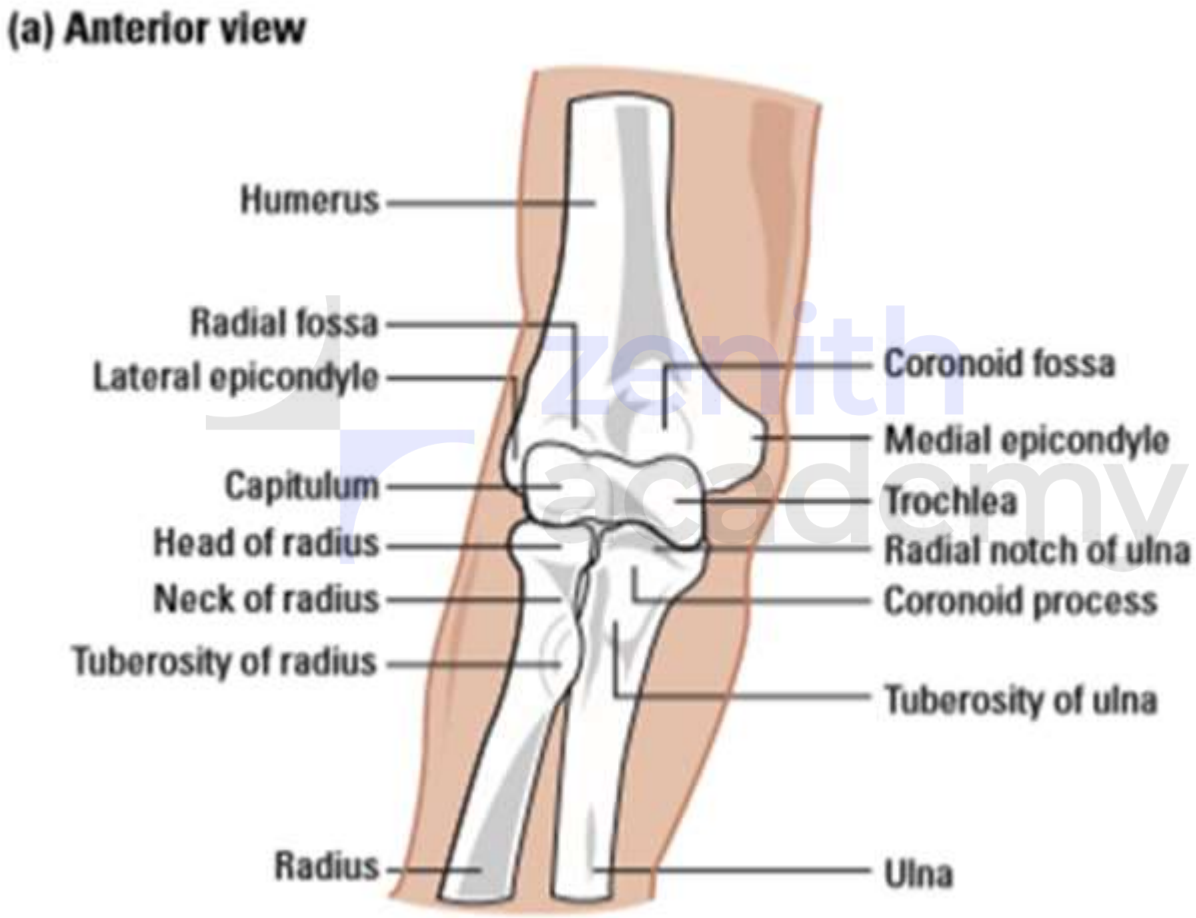
INTERNAL ROTATION



EXTERNAL ROTATION



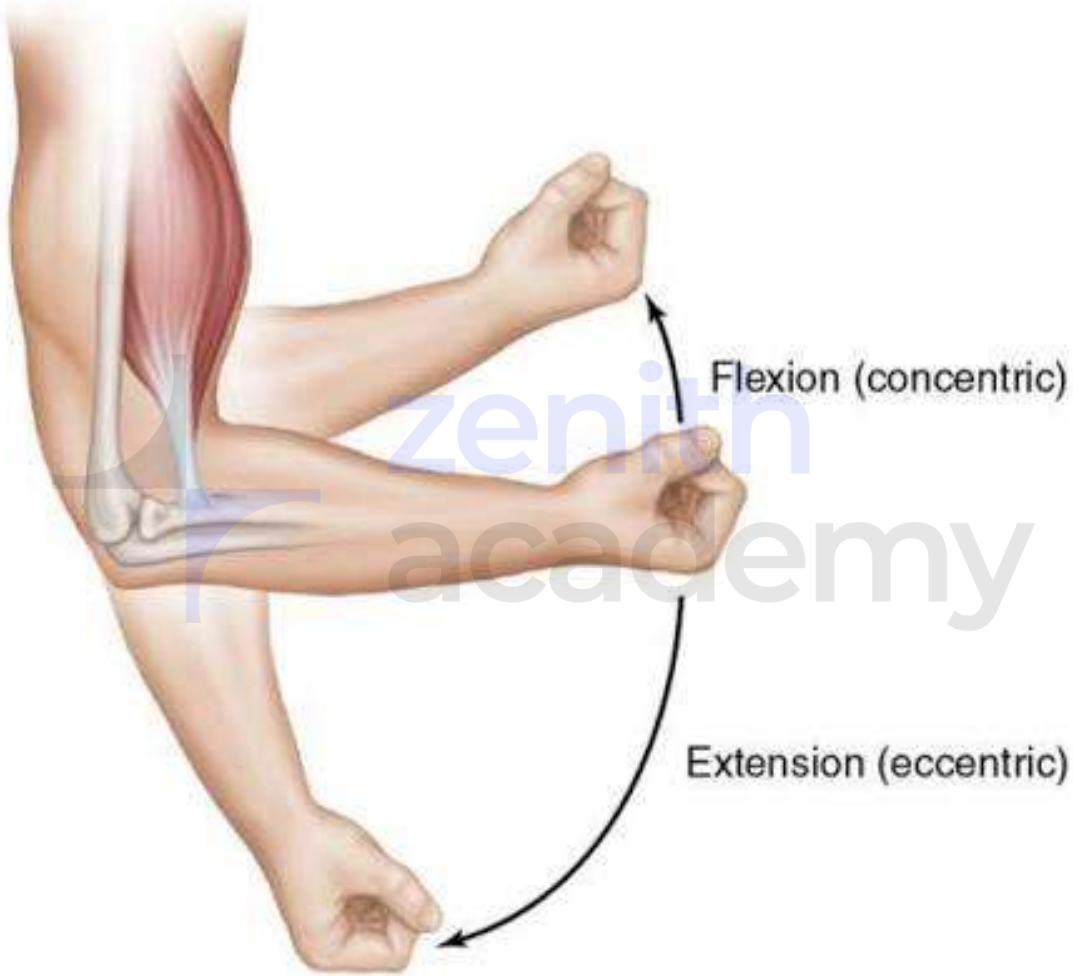
ELBOW JOINT



Hinge joint

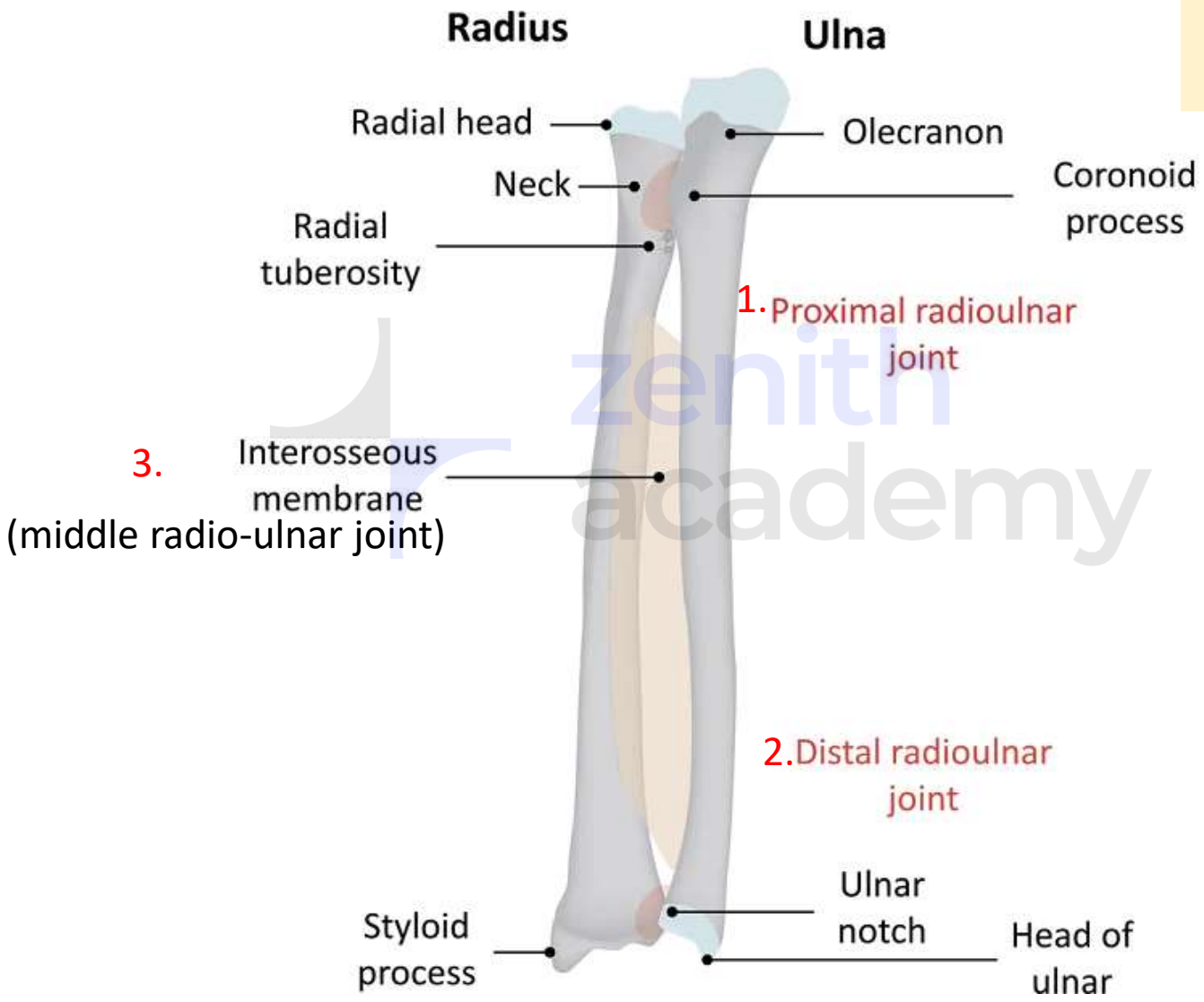
Movements

- Flexion
- Extension



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RADIO-ULNAR JOINT

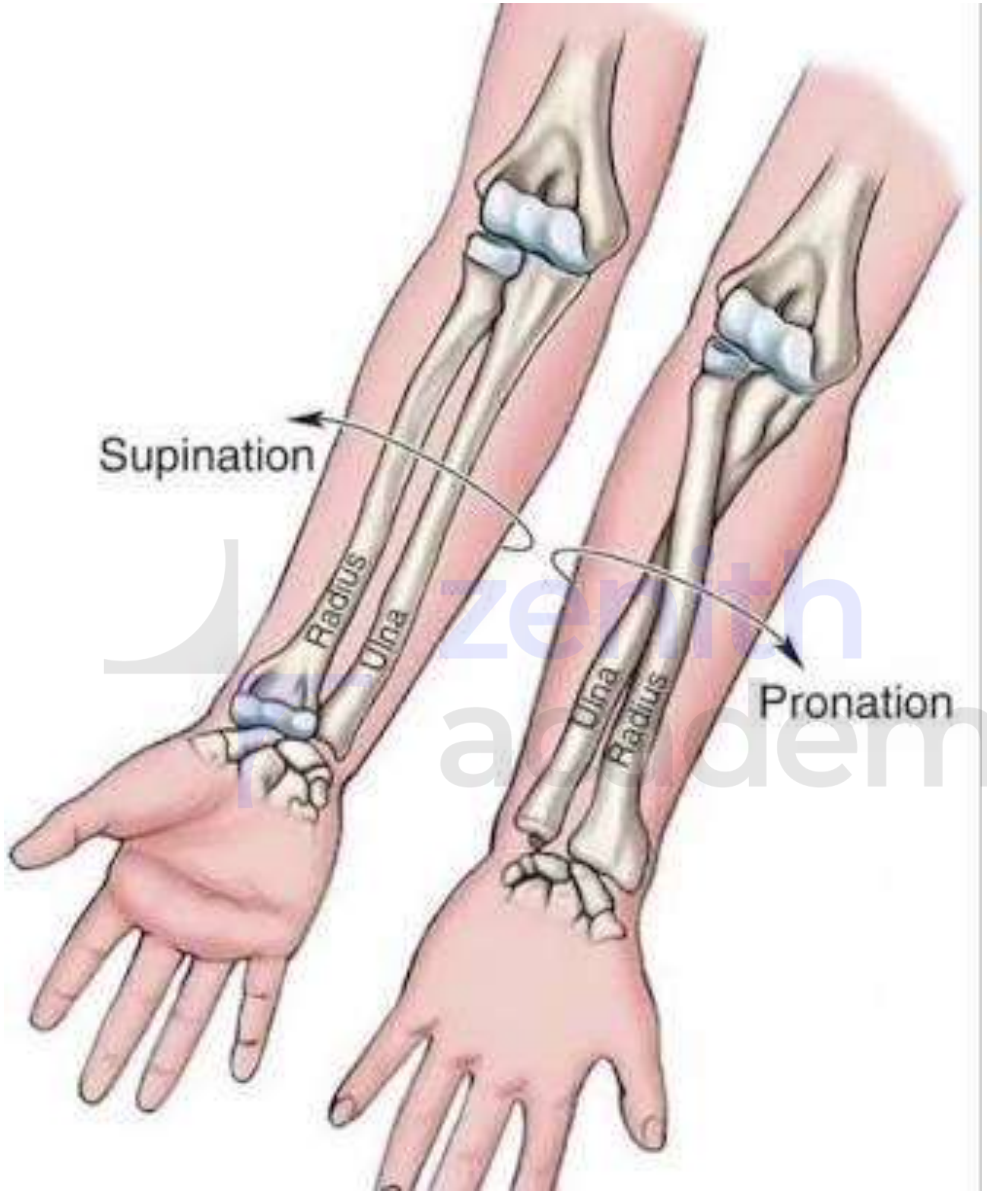


Pivot joint in upper and lower syndesmosis joint in middle

Movements

- Supination
- Pronation

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WRIST JOINT / RADIO-CARPAL JOINT

Condyloid joint

Movements

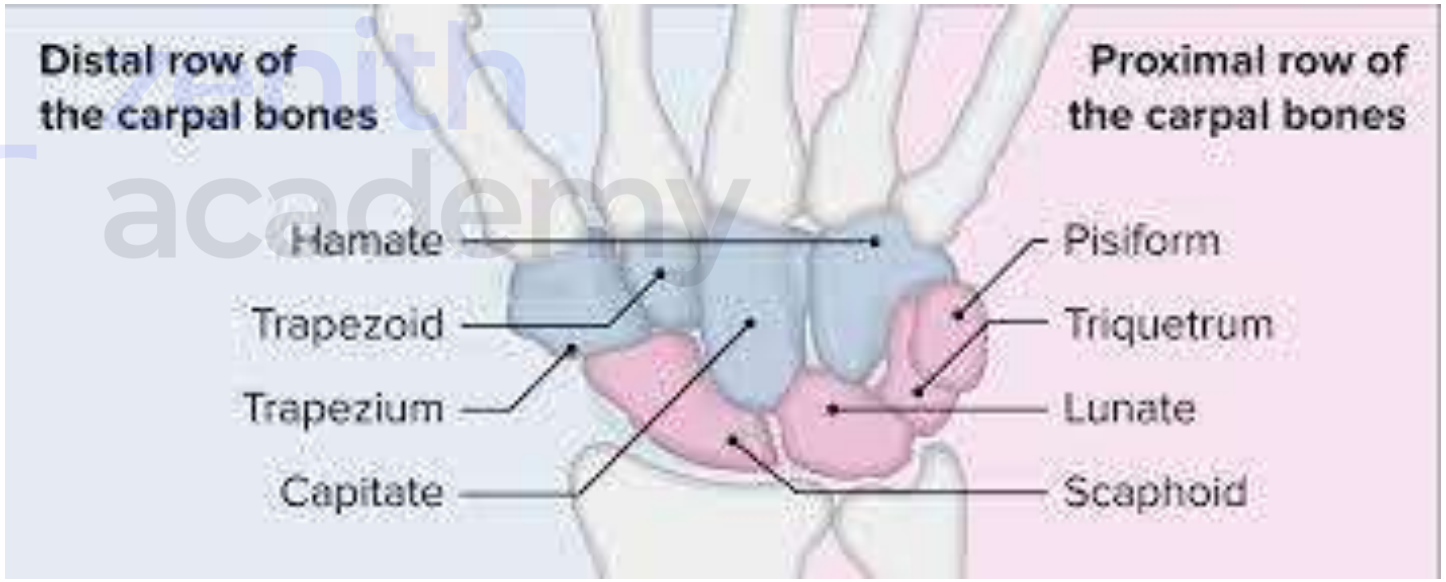
- Abduction
- Adduction
- Flexion
- Extension

CARPAL JOINTS

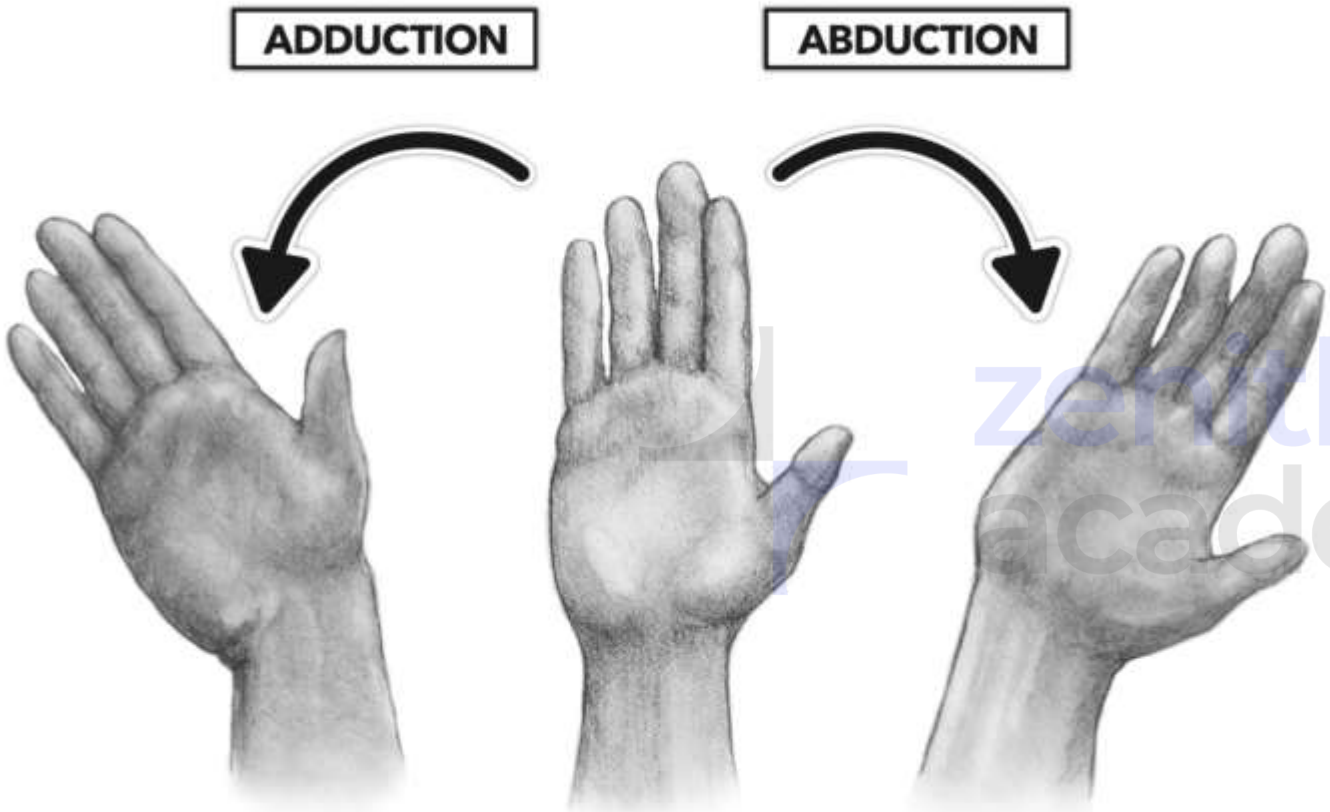
Gliding joint

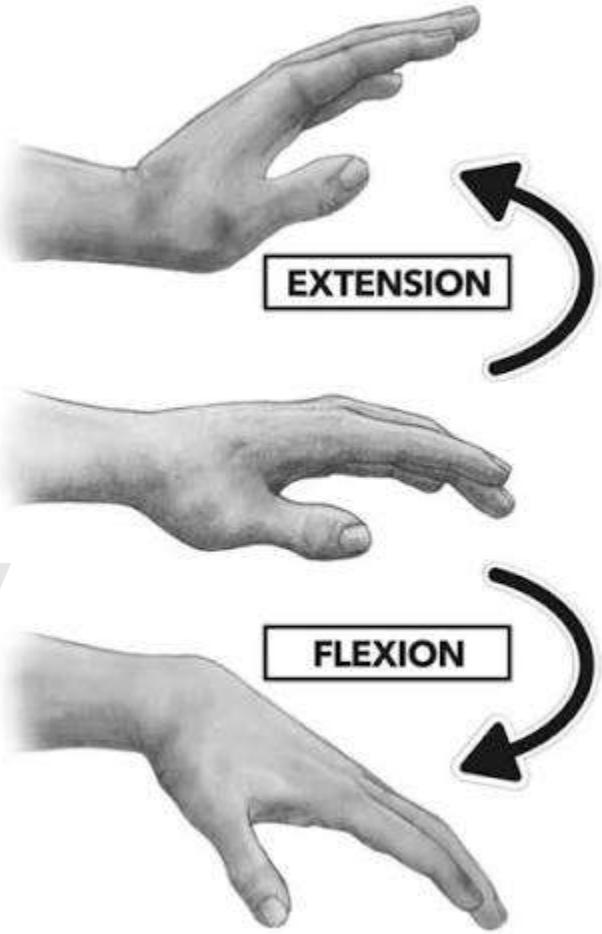
CARPO- METACARPAL JOINTS

Gliding joint



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META CARPO PHALANGEAL JOINT



Condyloid joint

Movements

- Abduction
- Adduction
- Flexion
- Extension

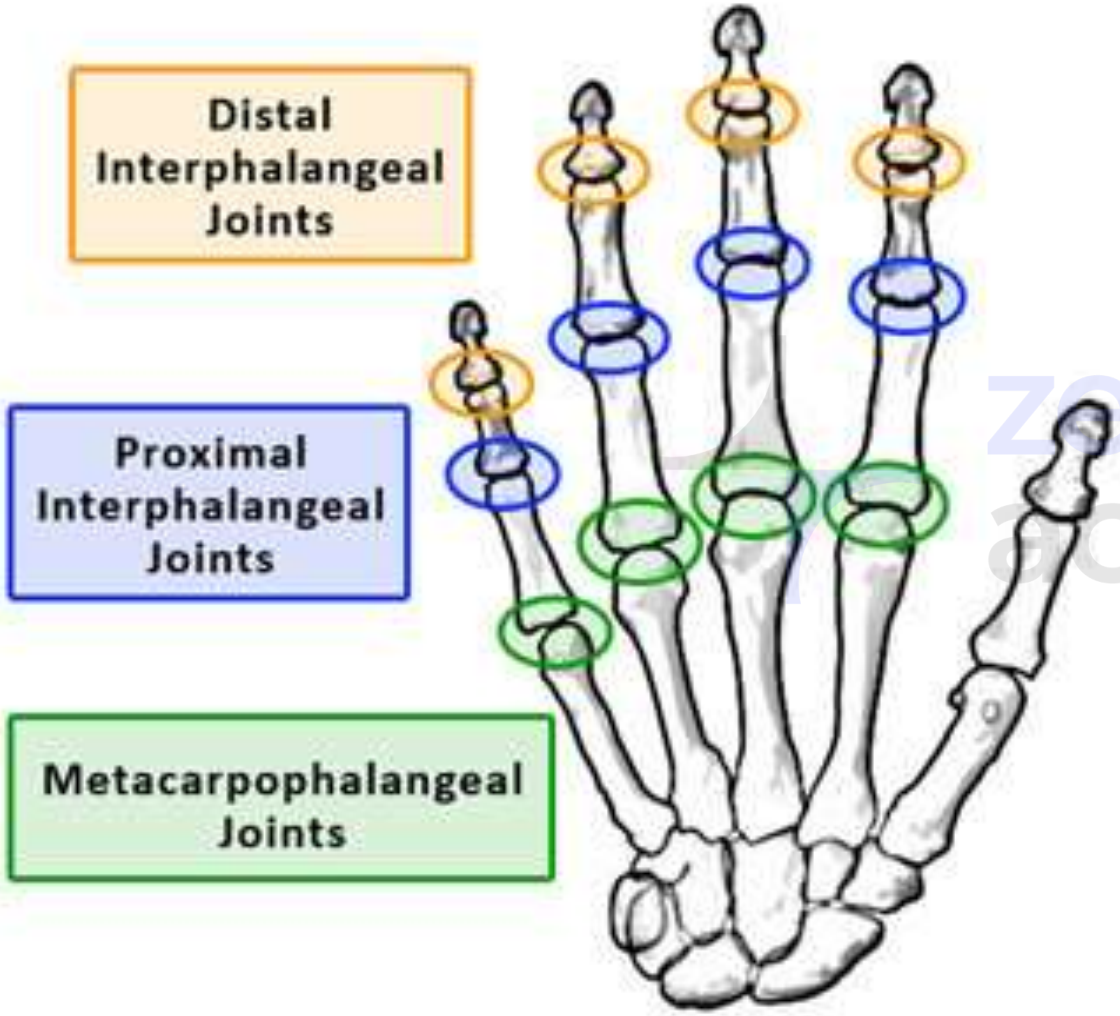


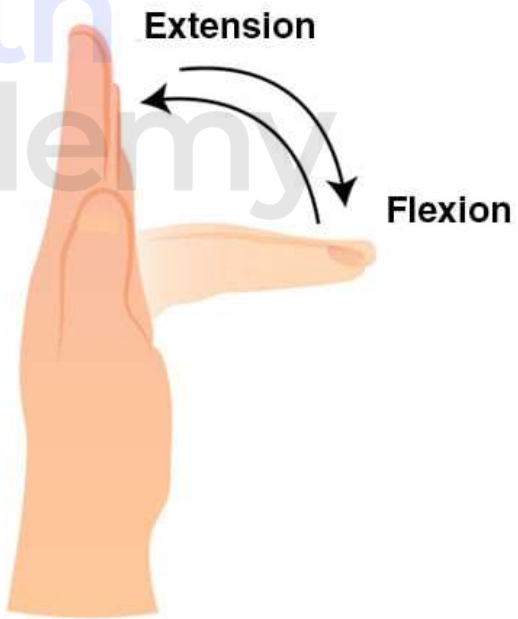
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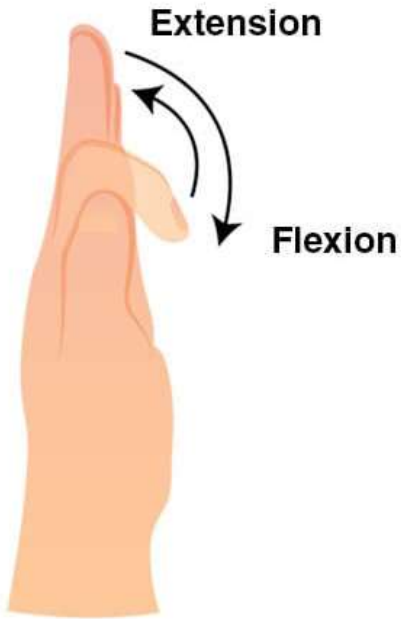
INTERPHALANGEAL JOINT

Hinge joint

Movements
- Flexion
- Extension







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1. Gomphosis is seen in:

a) Skull b) Teeth sockets c) Vertebrae d) Shoulder

2. Syndesmosis joint example:

a) Skull b) Radio-ulnar c) Shoulder d) Knee

3. Intervertebral joints are:

a) Fibrous b) Synovial c) Cartilaginous d) Pivot

4. Joint cavity is present in:

a) Fibrous b) Cartilaginous c) Synovial d) Suture

5. Gliding joints allow:

a) Rotation b) Sliding c) Flexion d) Extension

1-B
2-B
3-C
4-C
5-B

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6. Ball and socket joint allows:

a) One plane b) Two planes c) No movement d) All directions

7. Hinge joint allows:

a) Rotation b) Flexion and extension c) Gliding d) Circumduction

8. Pivot joint allows:

a) Rotation b) Flexion c) Extension d) Abduction

9. Adduction means movement:

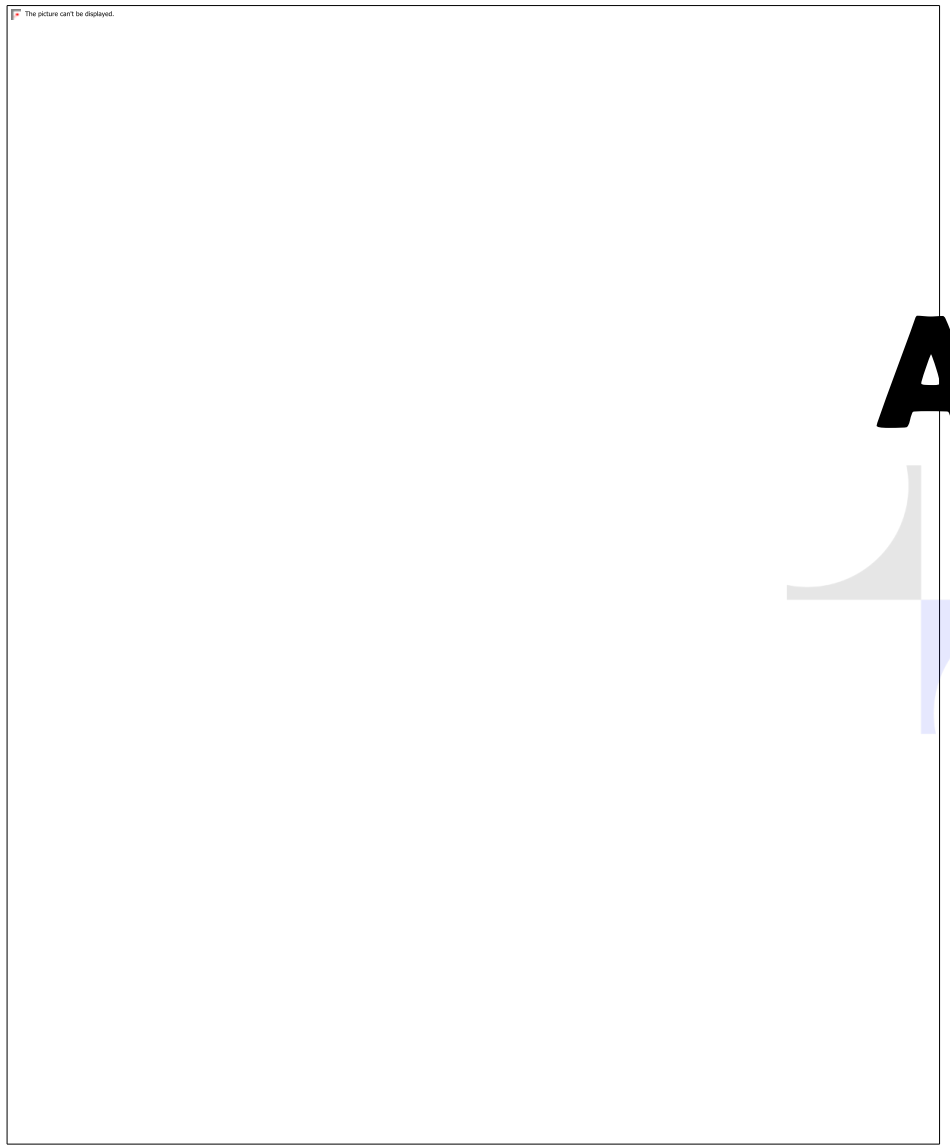
a) Away b) Towards midline c) Rotation d) Extension

10. Sternoclavicular joint is:

a) Hinge b) Gliding c) Pivot d) Saddle

6-D
7-B
8-A
9-B
10-B

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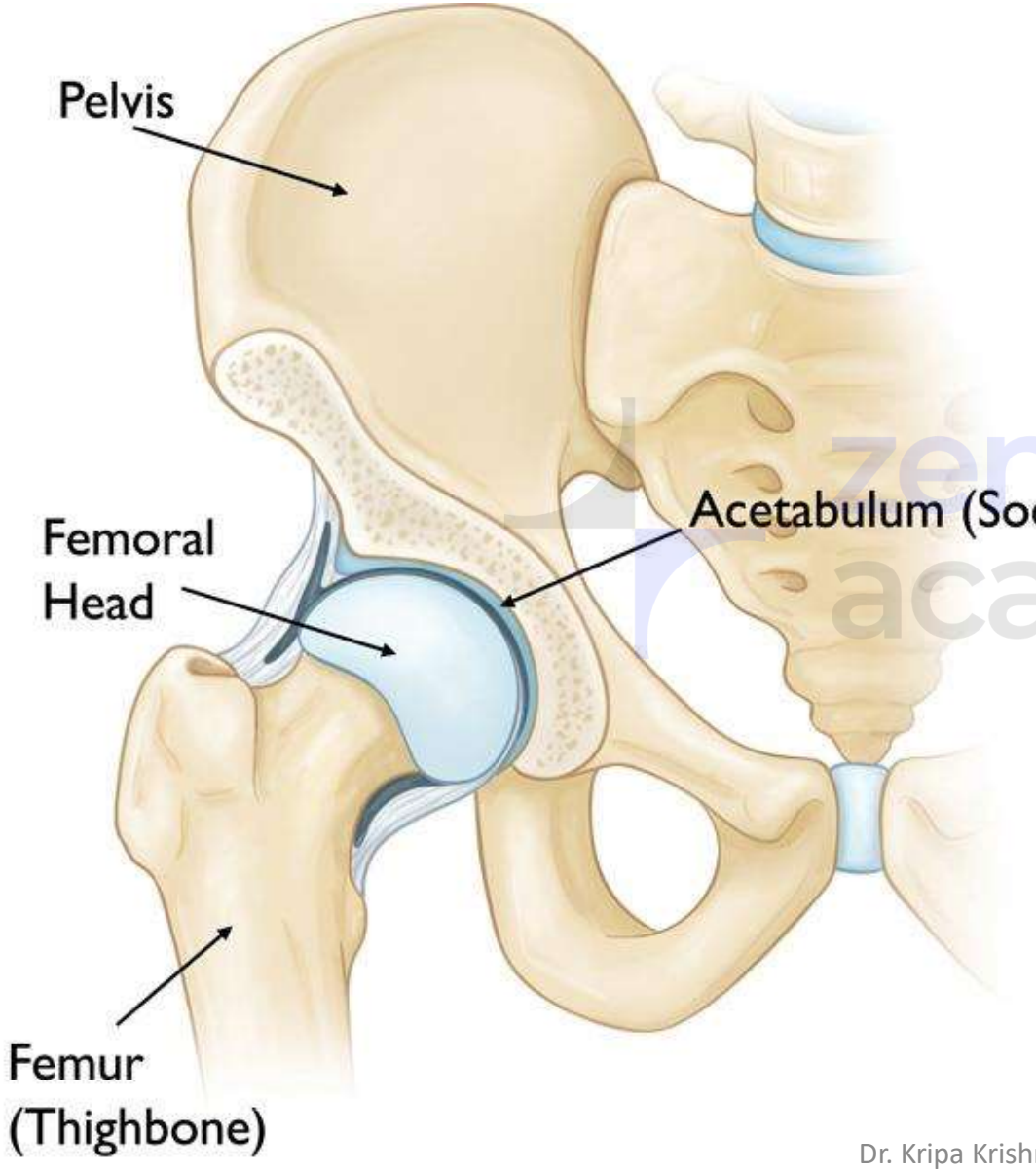
Articular system

Part-II

Dr. Kripa Krishnan

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HIP JOINT

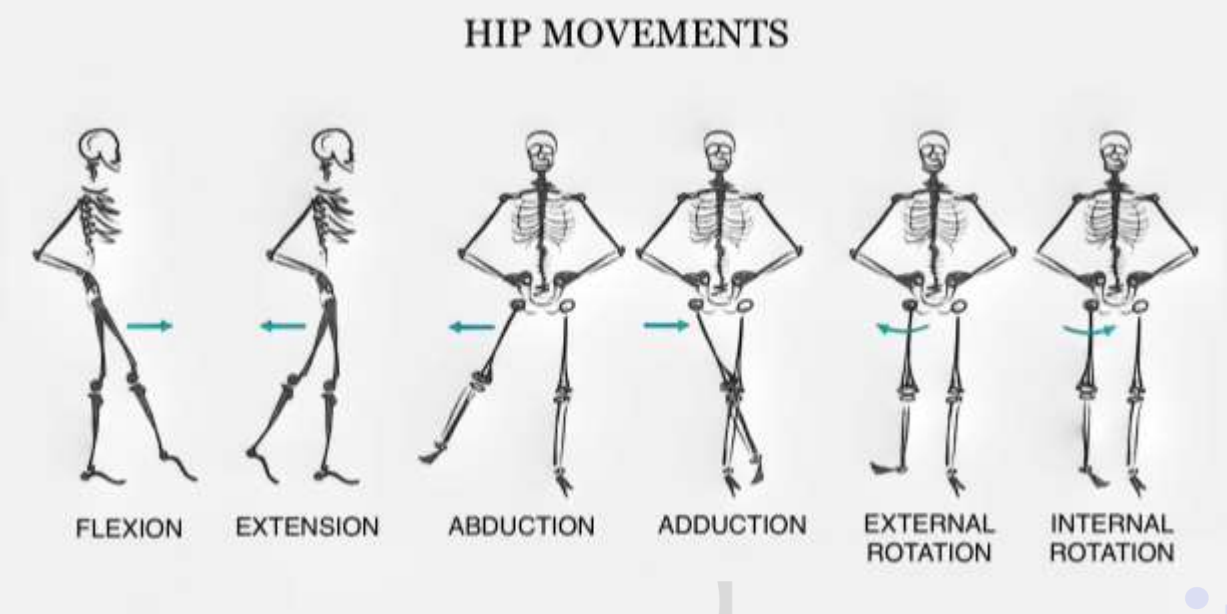


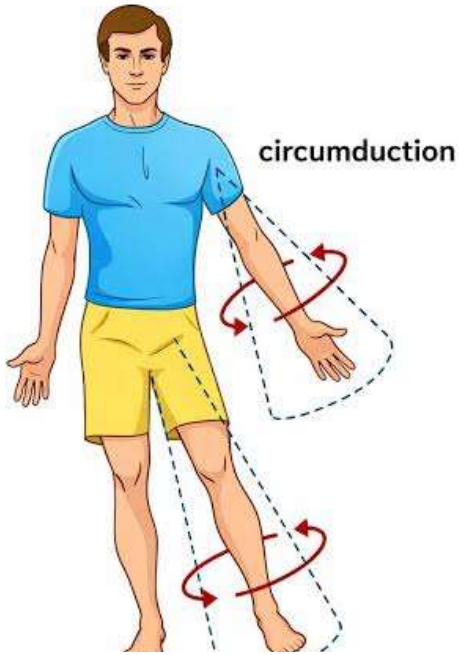
Ball and socket joint

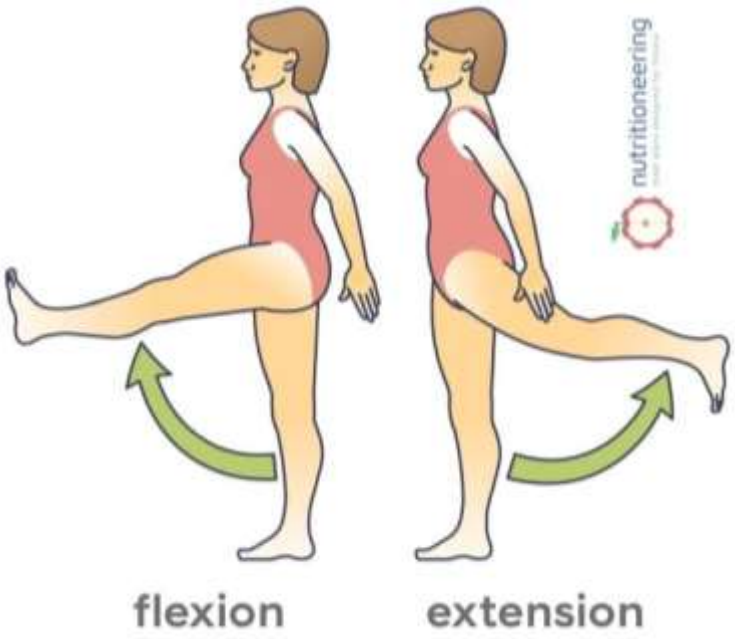
Movements

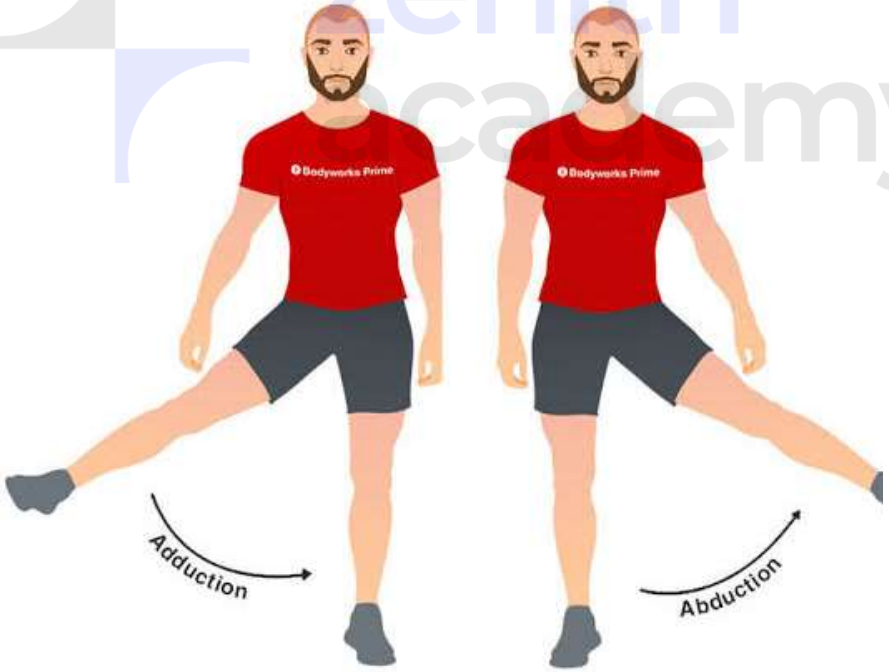
- Flexion
- Extension
- Abduction
- Adduction
- Medial rotation
- Lateral rotation
- Circumduction

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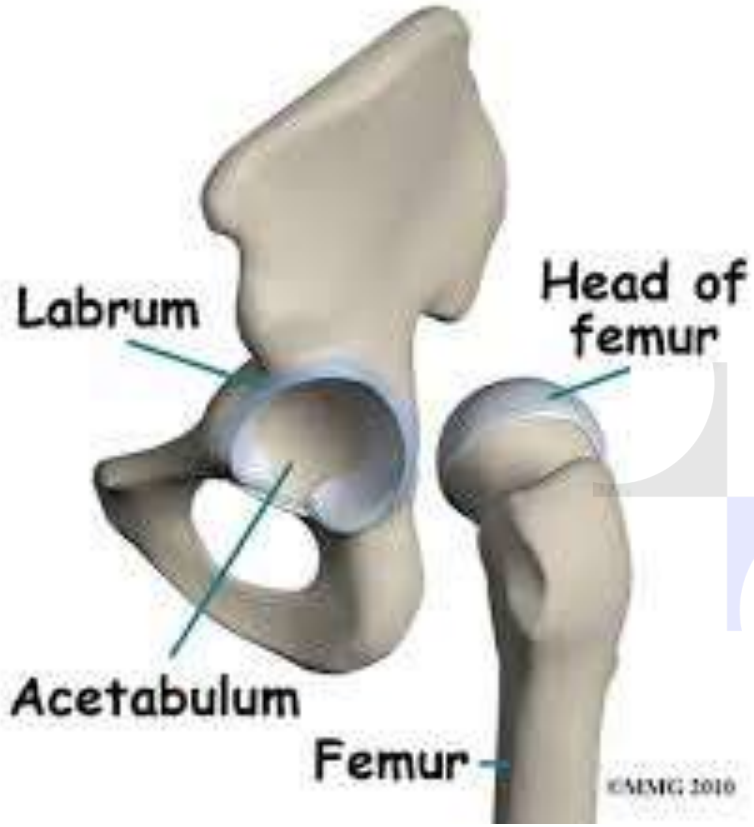


Lateral rotation



Medial rotation

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Acetabular labrum

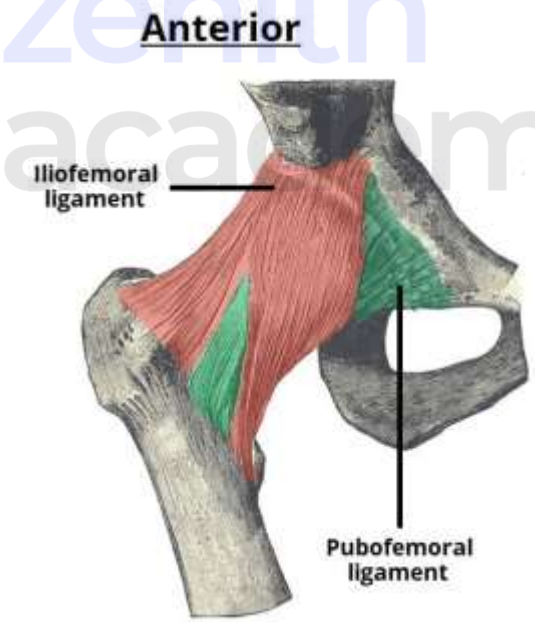
A specialized ring of fibrocartilage that lines the outer rim of the hip socket (acetabulum). Its primary roles are to deepen the socket to increase joint stability.

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Capsular Ligament (Joint Capsule)

- Think & strong ligament limits the movement of joint in all directions.
- The ligament is specially strengthened by bands of fibres.
- One of the most important of these bands lies in front of the joint, the ilio- femoral ligament.
- It limits extension at the joint & helps to maintain the erect position in standing.

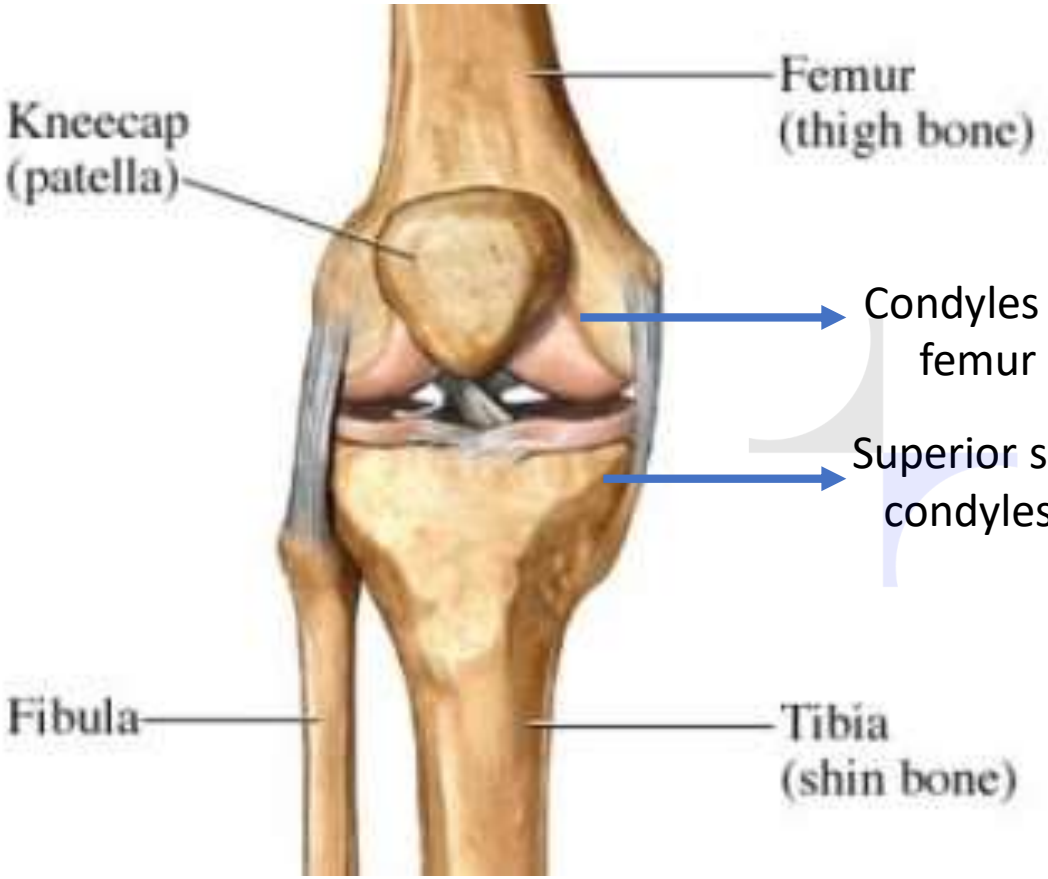






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KNEE JOINT

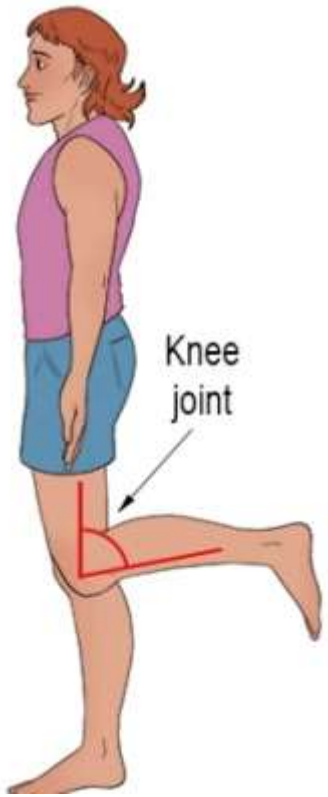


Modified hinge joint
(it also permits slight rotation that a simple hinge cannot)

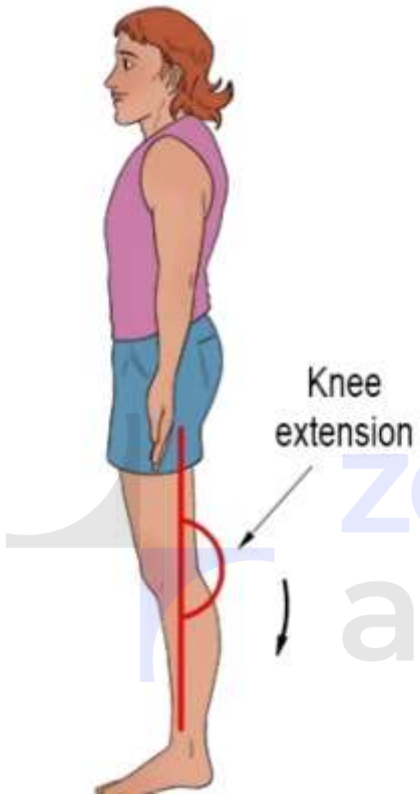
Movements

- Flexion
- Extension
- Slight medial rotation

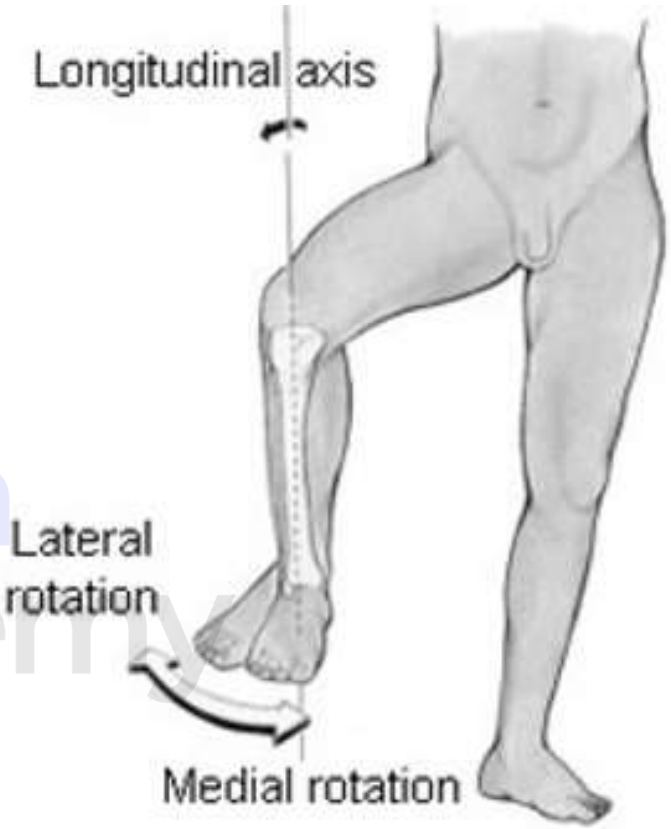
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Flexion



Extension

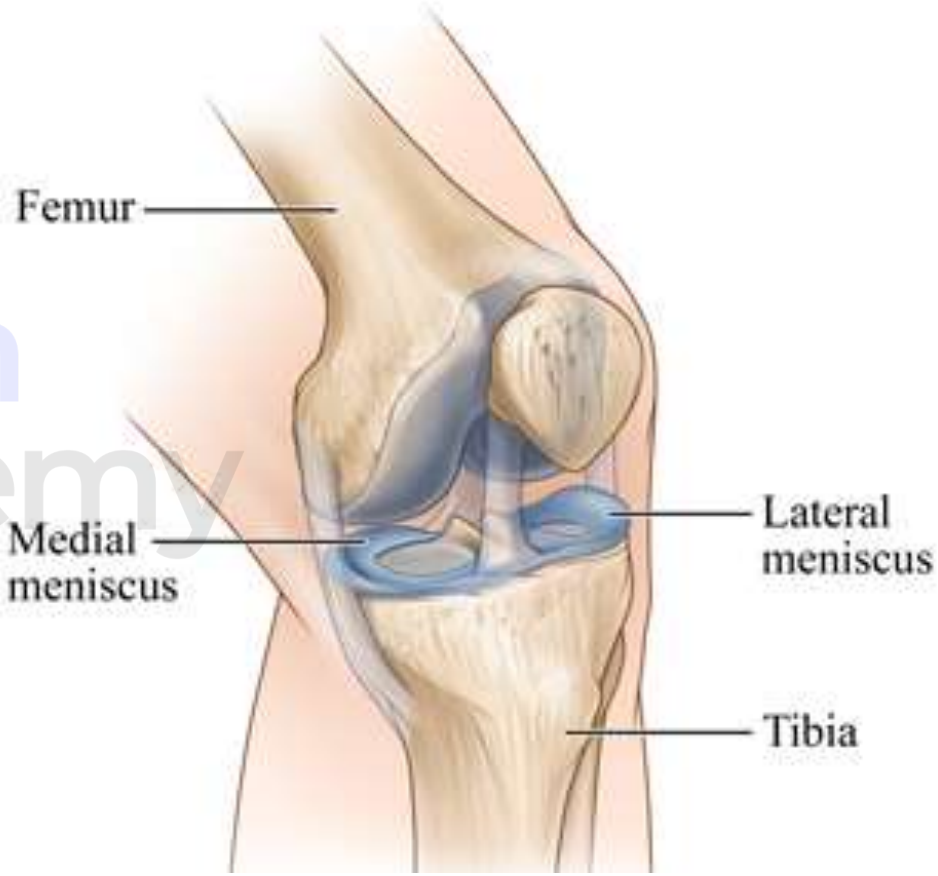


Is a rotational movement that can only occur when the knee joint is bent.

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Semilunar cartilages (menisci)

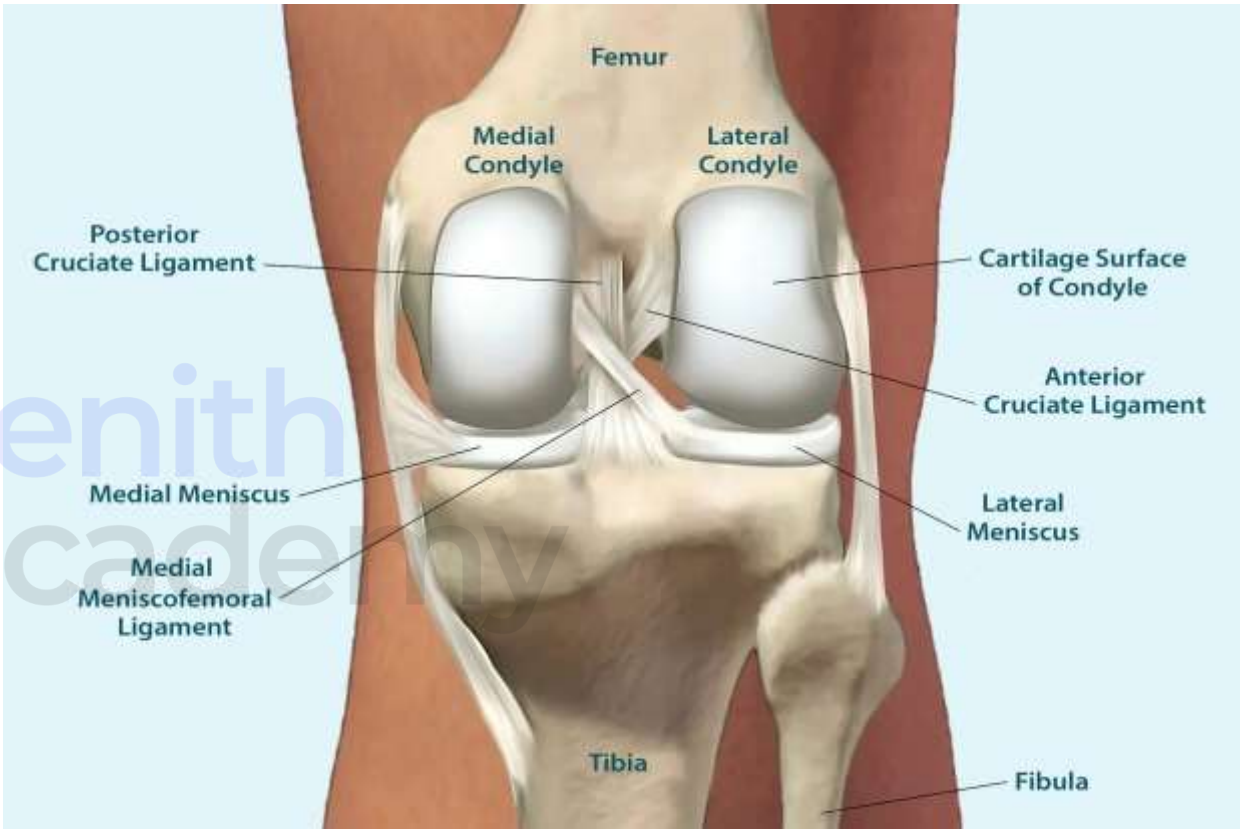
Two crescent-shaped pads of fibrocartilage placed on plateu like articular surface of tibia, they act as critical "shock absorbers.



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Cruciate ligaments

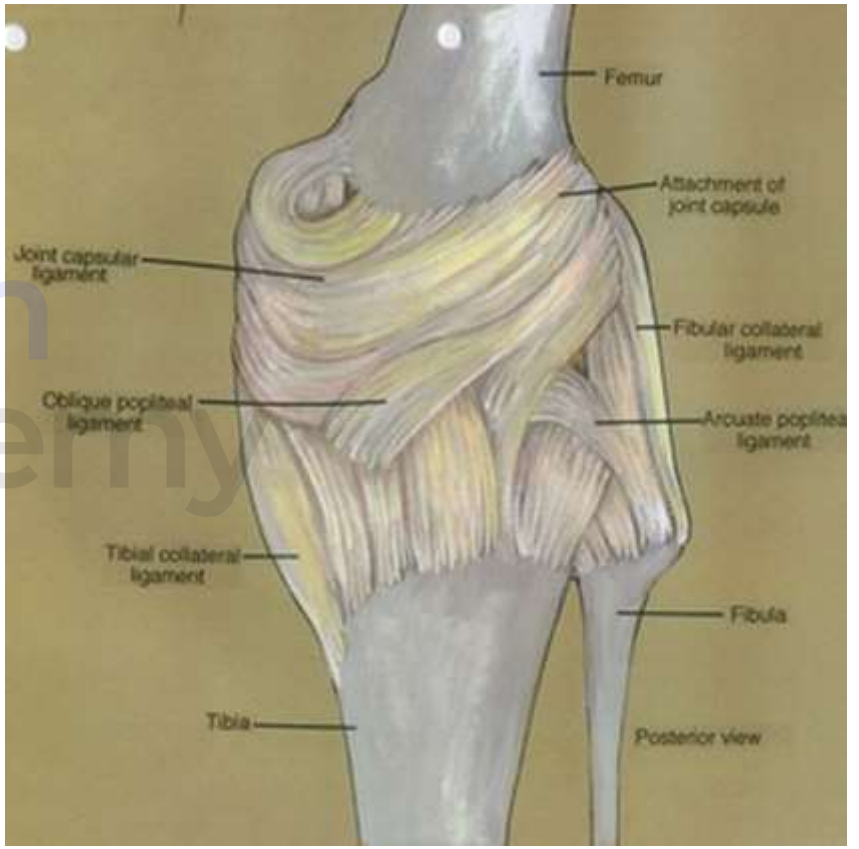
- Pass from the top of tibial condyles to the intercondyloid notch of femur.
- These serve to limit the movement of the knee joint and bind the bones more firmly together.



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Capsular ligaments
(articular capsule)

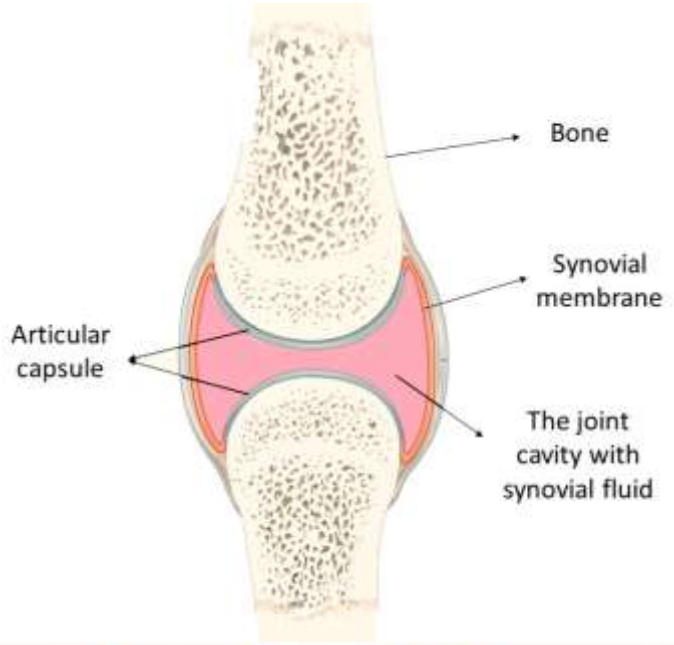
- A dense, fibrous envelope that surrounds the knee joint, connecting the femur to the tibia.
- It acts as a protective, stabilizing container, reinforced by surrounding tendons and ligaments to maintain joint integrity while sealing in synovial fluid.

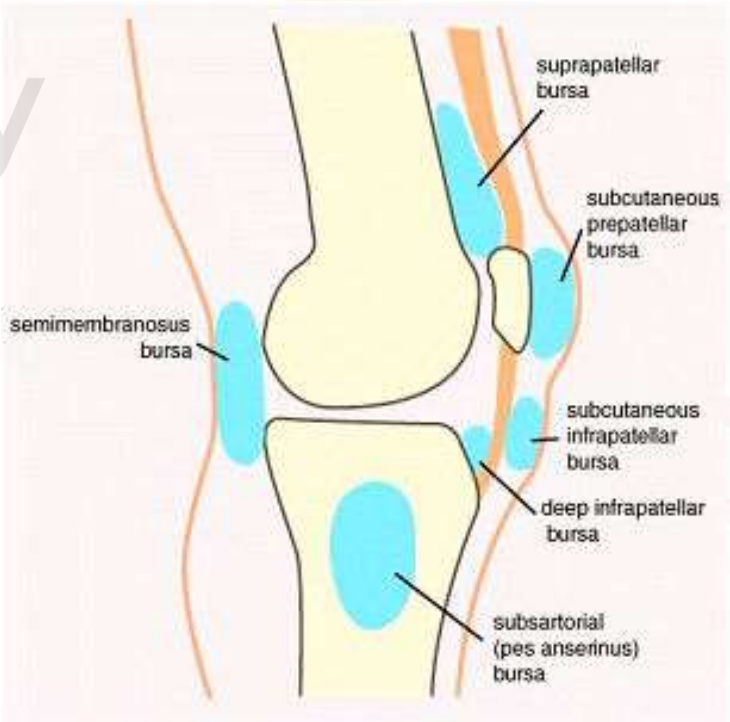


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Synovial membrane

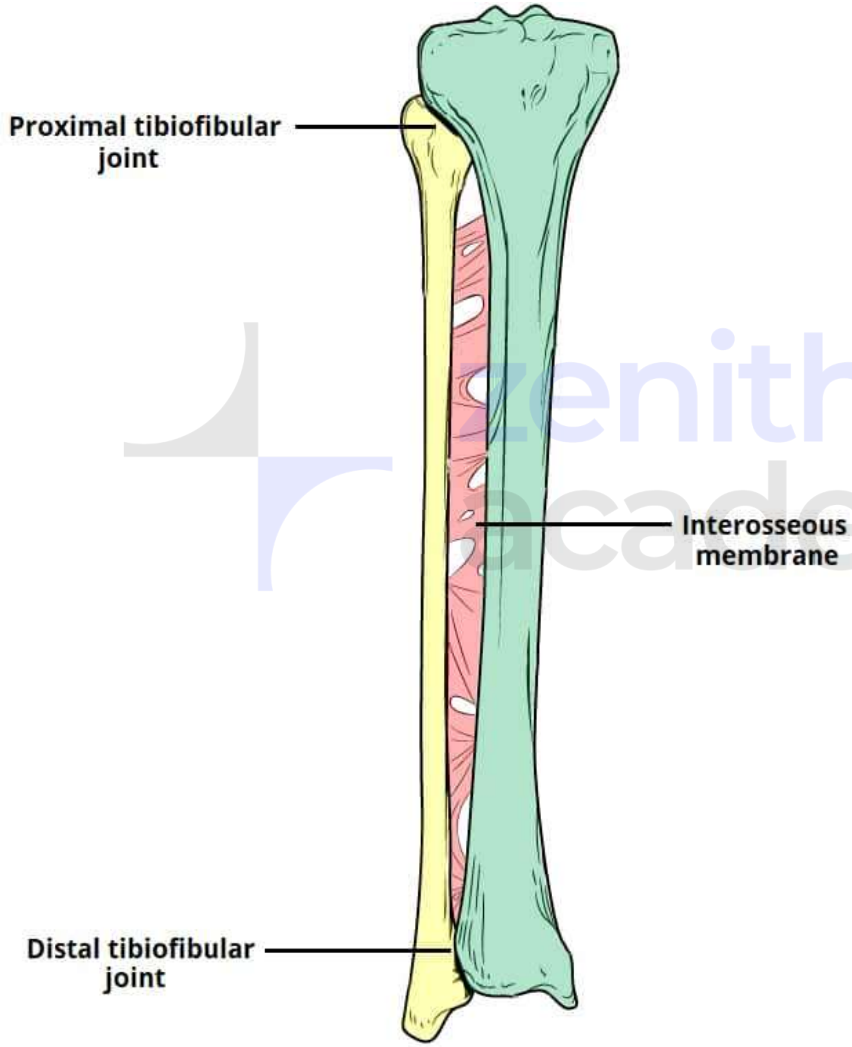
- The knee joint contains the largest synovial membrane in the body.
- In addition to lining the joint structures, it extends upwards and downwards beneath the ligaments of the patella and forms several bursae .
- Bursae- small, synovial fluid-filled sacs that reduce friction between tendons, ligaments, and bones, allowing for smooth joint movement.





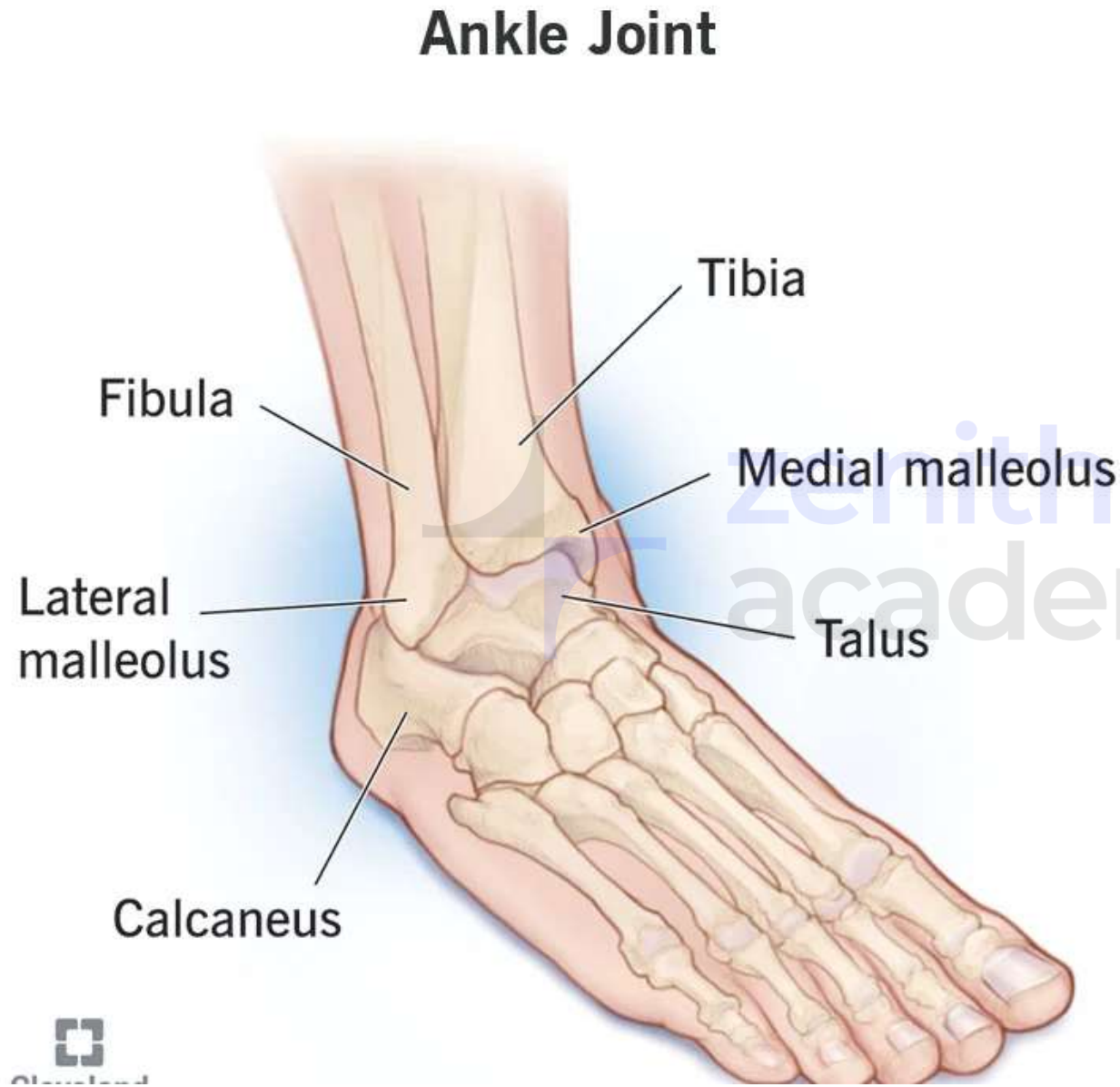
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TIBIO-FIBULAR JOINT



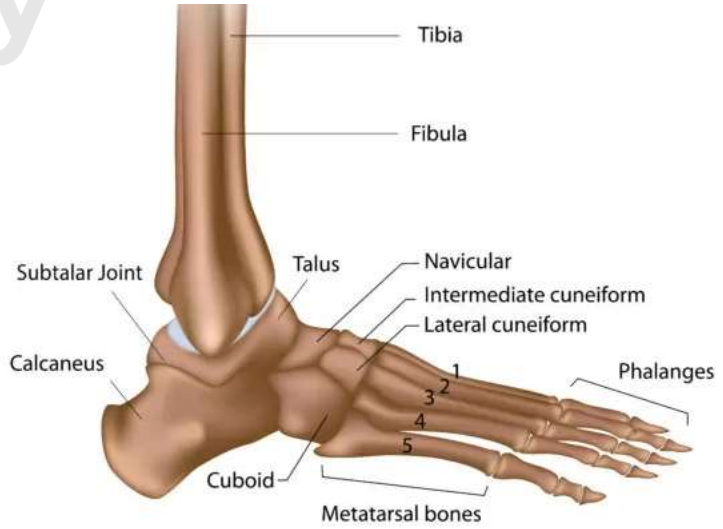
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Hinge joint



Movements

- Abduction
- Adduction
- Dorsi-flexion
- Plantar-flexion



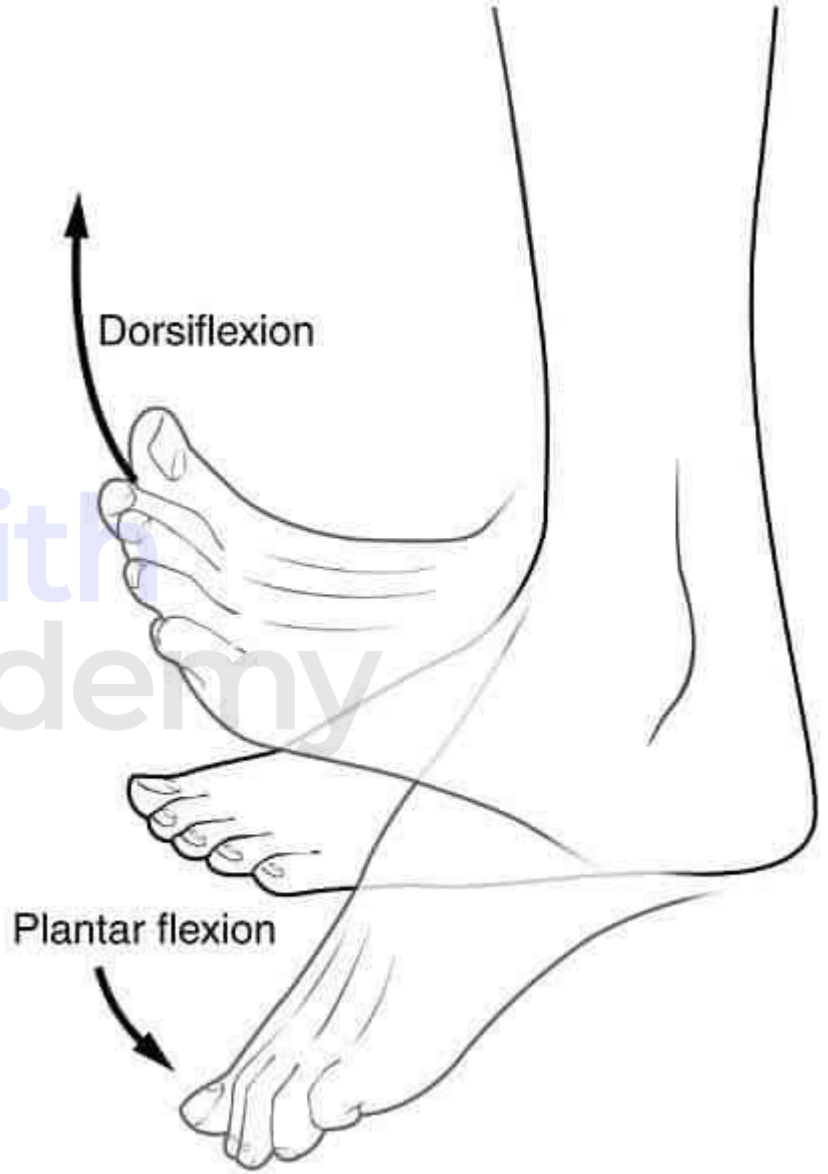
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Adduction



Abduction



Inversion



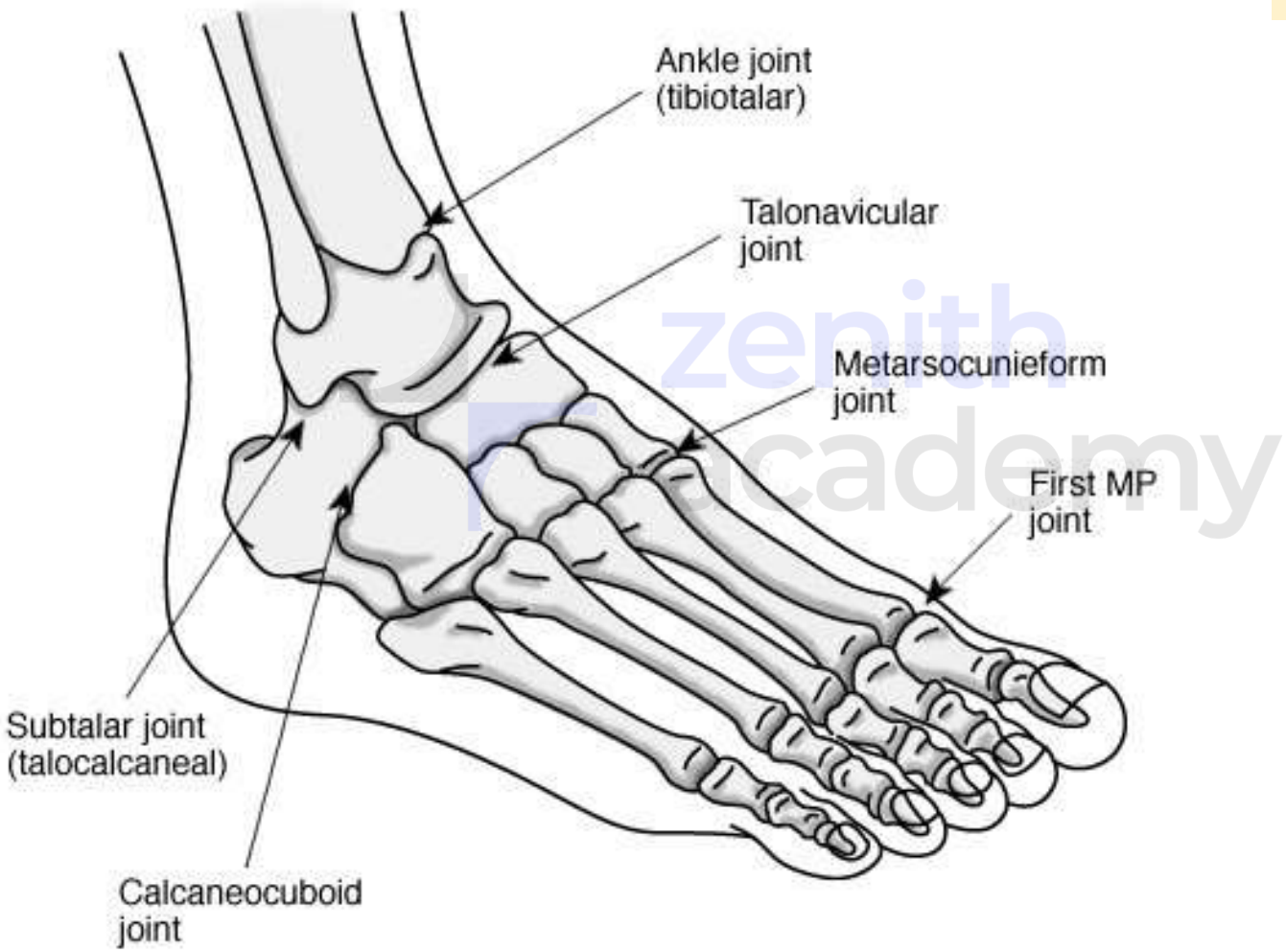
Eversion



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TARSAL JOINT

Gliding



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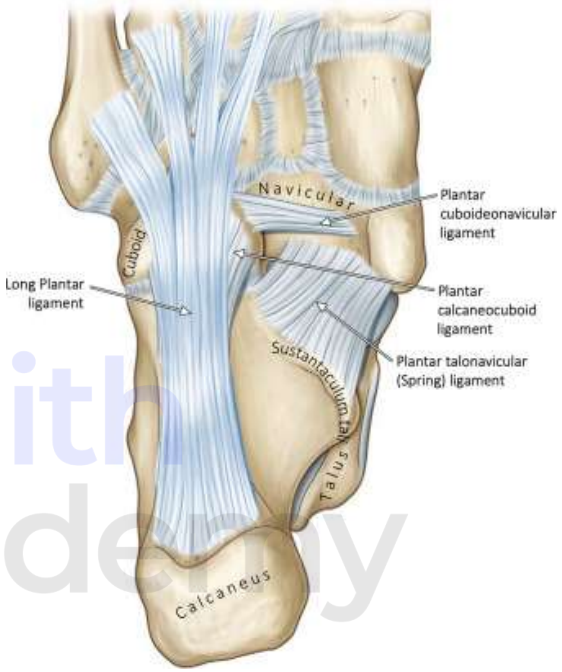
The bones are united by ligaments

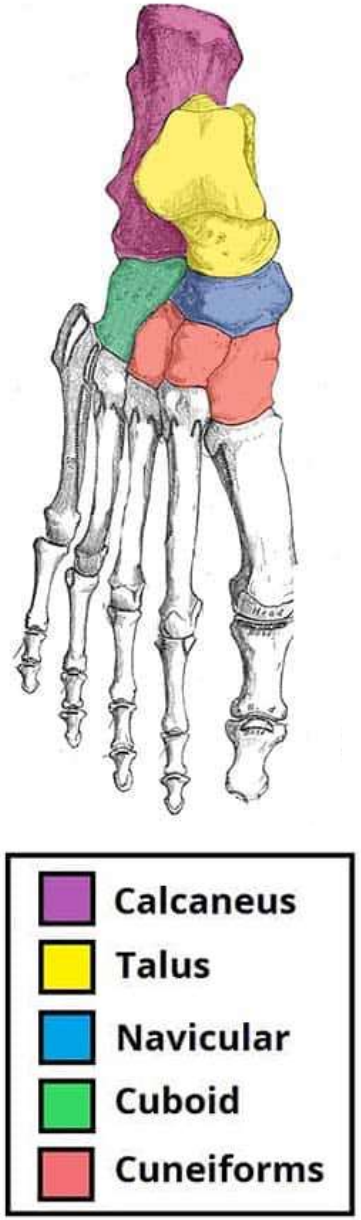
Dorsal Tarsal Ligaments: Thin, flat bands covering the dorsum of the joints

(Dorsal- the upper or top surface of the foot)

Plantar Tarsal Ligaments: Stronger bands that support the longitudinal arch.

Interosseous Tarsal Ligaments: Strong fibers connecting adjacent tarsal bones within the foot.



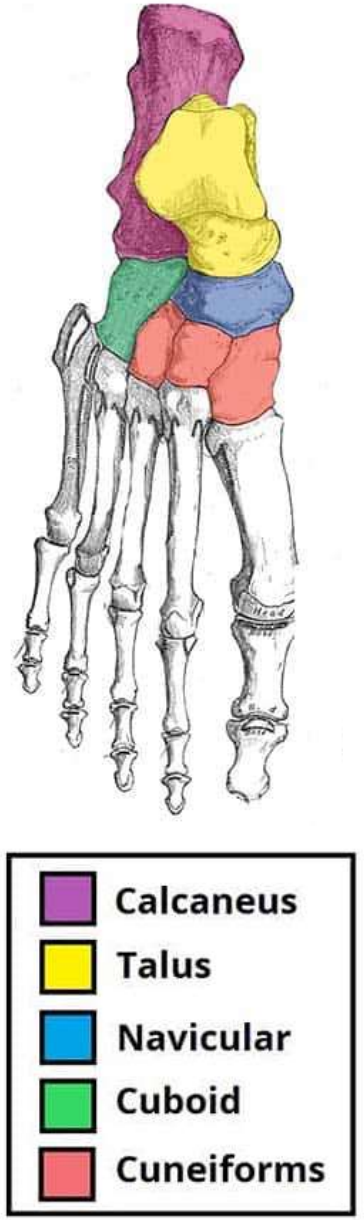


Eg:- Connection between calcaneum and talus – interosseous ligament

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Movements

- Little rocking occurs at talo-calcaneal joint, which is similar to adduction and abduction.
- Joint between head of talus & navicular — Medio-tarsal joint
  calcaneum& cuboid — Sub taloid joint
- At these joints inversion & eversion take place
- In conjunction with inversion & eversion, slight abduction and adduction occur at talo-calcaneal joint



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Adduction



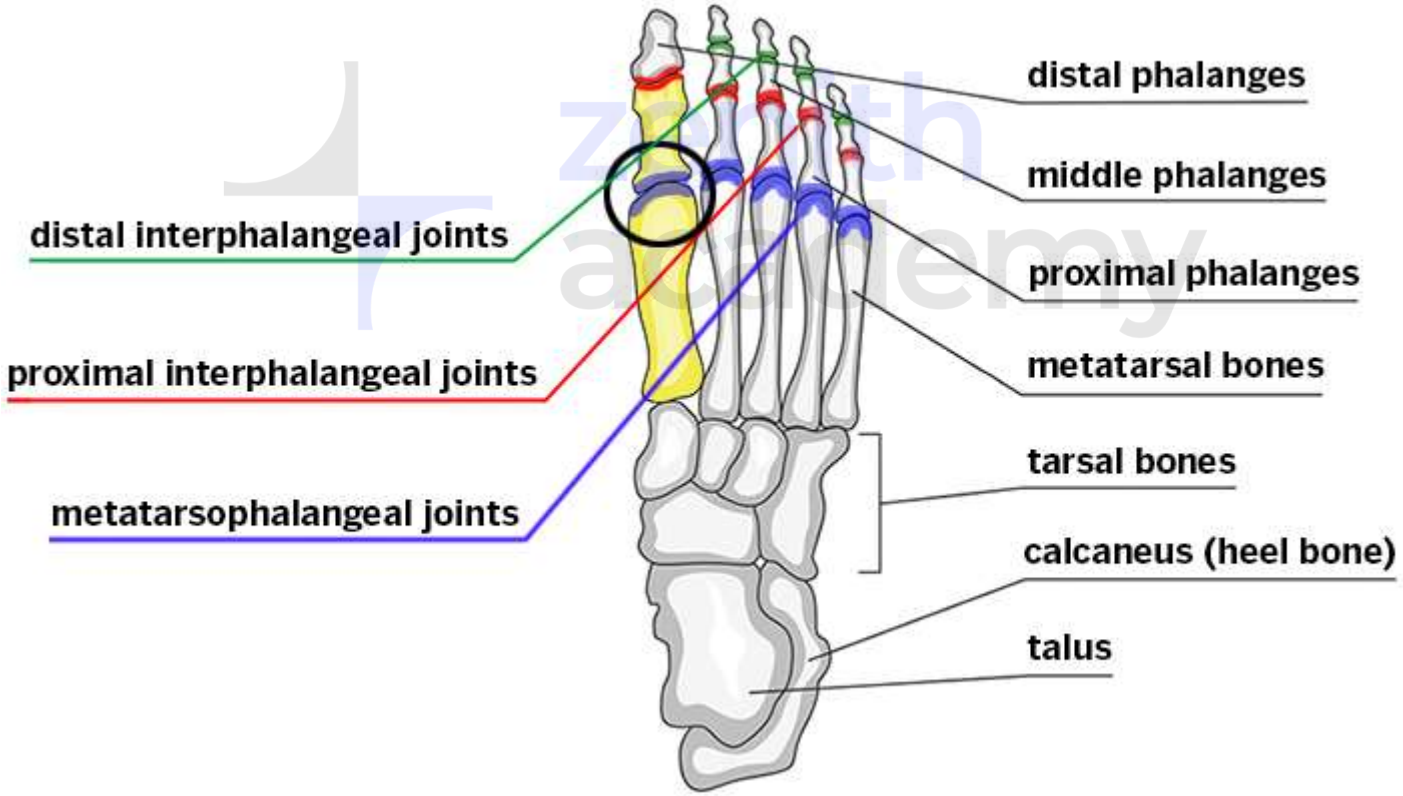
Abduction





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TARSO-METATARSAL - Gliding joint
METATARSO-PHALANGEAL – Condyloid joint
INTERPHALANGEAL JOINT- Hinge joint



1. The hip joint is a:

   a) Hinge joint b) Pivot joint c) Ball and socket joint d) Gliding joint

2. Which structure deepens the hip socket?

   a) Meniscus b) Acetabular labrum c) Bursa d) Synovial membrane

3. The hip joint helps maintain erect posture mainly due to:

   a) Synovial membrane b) Capsule c) Iliofemoral ligament d) Menisci

4. The knee joint is a:

   a) Simple hinge joint b) Pivot joint c) Saddle joint d) Modified hinge joint

5. Menisci in the knee act as:

   a) Ligaments b) Shock absorbers c) Muscles d) Tendons

1-c
2-b
3-c
4-d
5-b

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6. Cruciate ligaments connect:

a) Femur to patella b) Tibia to fibula c) Tibia to femur d) Fibula to femur

7. The largest synovial membrane is found in:

a) Knee joint b) Hip joint c) Ankle joint d) Tarsal joint

8. Bursae function to:

a) Strengthen bones b) Reduce friction c) Produce cartilage d) Attach muscles

9. Syndesmosis type is seen in:

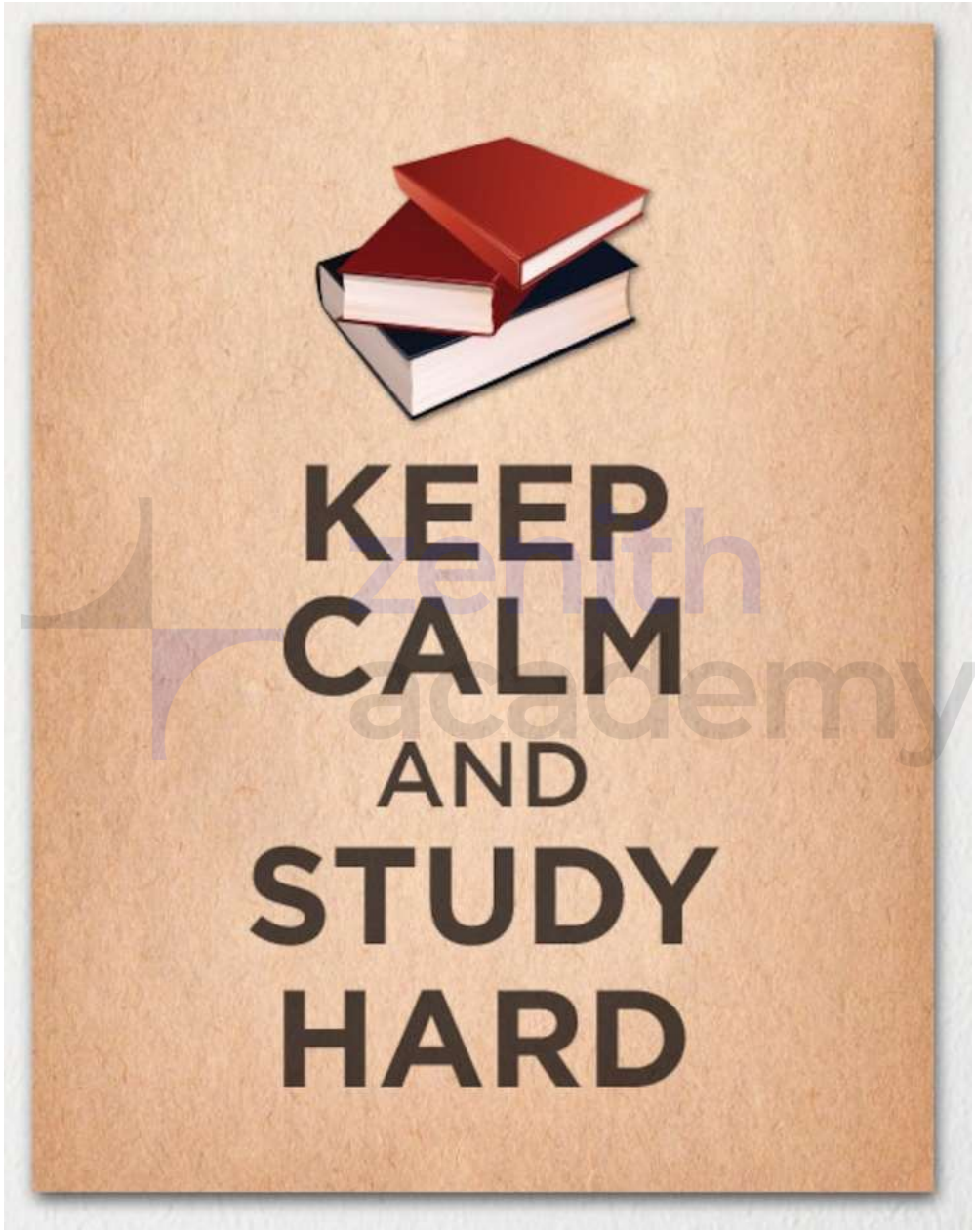
a) Hip joint b) Knee joint c) Tibiofibular joint d) Ankle joint

10. The ankle joint is a:

a) Pivot joint b) Saddle joint c) Hinge joint d) Condyloid joint

6- c
7-a
8-b
9-c
10-c

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